Centro de Investigación y de Estudios Avanzados del Instituto
Politécnico Nacional

DEPARTAMENTO DE FÍSICA

TRABAJO FIN DE GRADO

Sensibilidades a radio de carga de neutrinos en experimentos de neutrinos de reactor involucrando CEνNS

Realizado por
Mario Antonio Guerra Ruiz

Para la obtención del título de

Maestra en Ciencias
en la Especialidad de Física

Dirigido por
Dr. Omar Gustavo Miranda Romagnoli

Ciudad de México, Noviembre de 2026

Centro de Investigación y de Estudios Avanzados del Instituto
Politécnico Nacional

PHYSICS DEPARTMENT

Sensitivities to neutrino charge radius at reactor neutrino experiments involving CEνNS

Presented by
Mario Antonio Guerra Ruiz

In order to obtain the

Master of Science degree
speciality in Physics

Under the supervision of
PhD Omar Gustavo Miranda Romagnoli

Mexico City, November of 2026

Agradecimientos

agradecimiento aqui.

Resumen

El Modelo Estándar (ME) y la teoría electrodébil (EW, por sus siglas en inglés), así como las correcciones radiativas al ángulo de Weinberg ($\sin^2\theta_W$), predicen que los neutrinos, aunque eléctricamente neutros, poseen propiedades electromagnéticas cuantificables. Específicamente, una de estas propiedades es el radio de carga del neutrino (NCR, por sus siglas en inglés). Presentamos cómo diferentes configuraciones experimentales que involucran antineutrinos de reactor interactuando vía Dispersión Elástica Coherente Neutrino-Núcleo (CEνNS, por sus siglas en inglés) en argón líquido (LAr) pueden obtener resultados diferentes dependiendo de los valores de $\sin^2\theta_W$ y del NCR, cuantificando así la sensibilidad de dichos experimentos a estos parámetros del ME. Los resultados muestran la sensibilidad proyectada de experimentos basados en reactor cerca de la Ciudad de México para determinar el ángulo de Weinberg y el NCR del neutrino electrónico. Encontramos que la precisión proyectada en ambos parámetros escala de forma cercana a $1/\sqrt{N}$ con el tiempo de exposición y mejora en casi un orden de magnitud a lo largo de las masas de detector consideradas, de 10 a 200 kg, mientras que la inclusión de un factor de quenching realista para el argón, en lugar de una respuesta unitaria idealizada, degrada la sensibilidad a ambos parámetros por un factor de dos a tres debido a la pérdida de eventos de retroceso de baja energía por debajo del umbral. Incluso bajo este escenario realista, se proyecta que un detector de 200 kg expuesto durante 200 días restrinja $\sin^2\theta_W$ al nivel de 1-2% y el radio de carga del neutrino electrónico a $\langle r^2_{\nu_e}\rangle \sim 2\times10^{-33}$ cm^2, una precisión competitiva con las determinaciones de baja energía existentes del ángulo de mezcla débil y un objetivo concreto y comprobable para la estructura electromagnética del neutrino electrónico utilizando una fuente y una tecnología de detección ya disponibles a escala de un laboratorio nacional.

Abstract

Standard Model (SM) electroweak theory (EW) and radiative corrections to the Weinberg angle ($\sin^2\theta_W$) predict that neutrinos, while electrically neutral, possess quantifiable electromagnetic properties. Specifically, one such property is the neutrino charge radius (NCR). We present how different experimental configurations involving reactor antineutrinos interacting via Coherent Elastic Neutrino-Nucleus Scattering (CEνNS) in liquid argon (LAr) can yield differing results depending on the values of $\sin^2\theta_W$ and the NCR, thus quantifying the sensitivity of such experiments to these SM parameters. The results show the projected sensitivity of reactor-based experiments near Mexico City to determining the Weinberg angle and the electron-neutrino NCR. We find that the projected precision on both parameters scales close to $1/\sqrt{N}$ with exposure time and improves by nearly an order of magnitude across the detector masses considered, from 10 to 200 kg, while the inclusion of a realistic quenching factor for argon, rather than an idealized unit response, degrades the sensitivity to both parameters by a factor of two to three due to the loss of low-energy recoil events below threshold. Even under this realistic scenario, a 200 kg detector exposed for 200 days is projected to constrain $\sin^2\theta_W$ to the 1-2% level and the electron-neutrino charge radius to $\langle r^2_{\nu_e}\rangle \sim 2\times10^{-33}$ cm^2, a precision competitive with existing low-energy determinations of the weak mixing angle and a concrete, testable target for the electron-neutrino electromagnetic structure using a source and detector technology already available at a national laboratory scale.

Contents

1. Introduction

Particle physics is dedicated to unraveling the most fundamental constituents of the universe and the forces that govern their interactions. It explores the known elementary particles, such as quarks, leptons, and bosons, which constitute matter and mediate the fundamental forces, and it organizes them within the Standard Model (SM), the central theoretical framework of the field. The SM has been remarkably successful in describing the behavior of these particles, yet the discovery that neutrinos have mass and undergo flavor-changing oscillation showed that the model, in its original massless-neutrino formulation, is necessarily incomplete. This has made the neutrino sector one of the most active windows through which physics beyond the Standard Model (BSM) may first reveal itself.

Neutrinos were theoretically proposed by Pauli in 1930 to preserve energy and angular momentum conservation in beta decay, and they were experimentally discovered in 1956 by Cowan and Reines, after decades in which the technology required to detect their exceedingly weak interaction with matter simply did not exist. Since then, neutrinos have been studied in ever greater detail, from the discovery of their oscillations to the ongoing effort to pin down their absolute mass scale, their Dirac or Majorana nature, and their possible electromagnetic properties. Among these properties, the neutrino charge radius (NCR) occupies a particular place: unlike a magnetic moment or an electric dipole moment, it is not an independent addition to the SM but a genuine radiative prediction of electroweak theory itself, arising unavoidably at one loop even in the absence of any new physics. This makes the NCR simultaneously a precision test of the SM and a potential probe of BSM scenarios that would shift its predicted value away from the SM benchmark.

This work focuses on Coherent Elastic Neutrino-Nucleus Scattering (CEνNS), a process whose cross section is precisely predicted by the SM and is directly sensitive to the electroweak couplings that the NCR modifies. This sensitivity, however, comes with the fact that CEνNS events correspond to extremely low-energy nuclear recoils, which are experimentally challenging to observe and which must be related, through a detector-dependent quenching factor, to the ionization energy that is actually measured. Furthermore, the shift induced by the NCR on the effective weak coupling is, at leading order, degenerate with a shift in the weak mixing angle $\sin^2\theta_W$ itself, so that any realistic sensitivity study must treat these two SM quantities together rather than in isolation. In this thesis we study the projected sensitivity of a reactor-based CEνNS experiment using a liquid argon (LAr) target to both $\sin^2\theta_W$ and the electron neutrino charge radius $\langle r^2_{\nu_e}\rangle$, using as a concrete benchmark the SBC-ININ-LAr proposal exposed to the antineutrino flux of the TRIGA Mark III reactor. We show how the interplay between detector mass, exposure time, and the realistic treatment of the quenching response determines how well these two parameters can be constrained, and to what extent their mutual degeneracy can be resolved.

The thesis introduces, in Chapter 2, the electroweak theory of the SM, beginning with a brief history of the neutrino and its role as the catalyst for the proposal and eventual discovery of the weak nuclear force, followed by the unification of that force with electromagnetism into the electroweak theory that underlies all subsequent calculations in this work. In Chapter 3, we turn to radiative corrections within this framework, showing how the inclusion of one-loop diagrams to the effective neutrino-fermion interaction predicts the electromagnetic properties of the neutrino, and in particular the neutrino charge radius, and how this prediction modifies the SM parameters, most notably $\sin^2\theta_W$, that are probed experimentally. Chapter 4 introduces CEνNS itself as the neutral-current interaction through which these SM parameters are experimentally accessed, presenting its cross section and combining it with a model of the reactor antineutrino flux to compute the expected number of events for a broadly-described experimental design, a quantity that later chapters use to test the influence of different SM parameter values. Chapter 5 presents the projected sensitivity of the SBC-ININ-LAr experiment to $\sin^2\theta_W$ and $\langle r^2_{\nu_e}\rangle$, for a range of detector masses, exposure times, and quenching factor treatments, together with the corresponding $\Delta\chi^2$ confidence intervals. Finally, we give our conclusions in Chapter 6.

2. Electroweak theory in the Standard Model of particles

This chapter serves as a description of the foundation for the theoretical framework of this work. Modern particle physics is based on the mathematical modeling of particles as excitations of quantum fields and how these fields interact with each other. This chapter begins with a brief history of the neutrino followed by its importance as the catalyst for the proposal and eventual discovery of a fundamental nuclear force, followed by its unification with a previously well understood force to introduce the main theory by which further advancements in neutrino property studies can be made.

2.1. Introduction

The Standard Model of particles (SM) is the most currently utilized theory of interactions of subatomic particles. It describes all matter particles and forces between them, excluding gravity, as interactions between 12 fermions (6 quarks and 6 leptons) with their corresponding anti-fermions, 4 spin-1 force-carrying gauge bosons and 1 scalar boson as shown in Fig. 2.1. Within this framework exist 3 distinct forces: strong nuclear force, mediated by the gluon (g) boson; electromagnetic (EM) force, mediated by the photon (γ) boson; and weak nuclear force, mediated by the $W^{\pm}$ and Z bosons. This work pertains to the description of the properties of the only family of particles that present no electric or color charge known as neutrinos (ν), wherein the SM exist three known flavors (ν_e, ν_μ and ν_τ) corresponding to the latter half of (electro)weak isospin doublets formed with the known charged leptons (e^-, μ^- and τ^-). This chapter aims to describe neutrinos in the SM, the weak force and its origin as a part of a larger broken symmetry known as "Electroweak Theory" (EW).

2.2. Symmetries and fermion content

This section provides a description of the relevant particles studied in this work as well as how their proposal and subsequent discovery led to the first description of the weak nuclear force.

2.2.1 Neutrinos

Neutrinos are charge-less, color-less, spin-$\frac{1}{2}$ particles with negligible masses compared to other fundamental particles first proposed in 1930 by Pauli W. [1], and later detected by C. L. Cowan et al. in 1956 [2], to solve known apparent contradictions in conserved quantities of beta decays. Beta decay was known to be as the emission of beta particles (electrons) by the nuclei of unstable atoms, wherein the difference in mass of the original and resulting nuclei of the decay should fix the energy at which the emitted particles are detected. Also of note, due to conservation

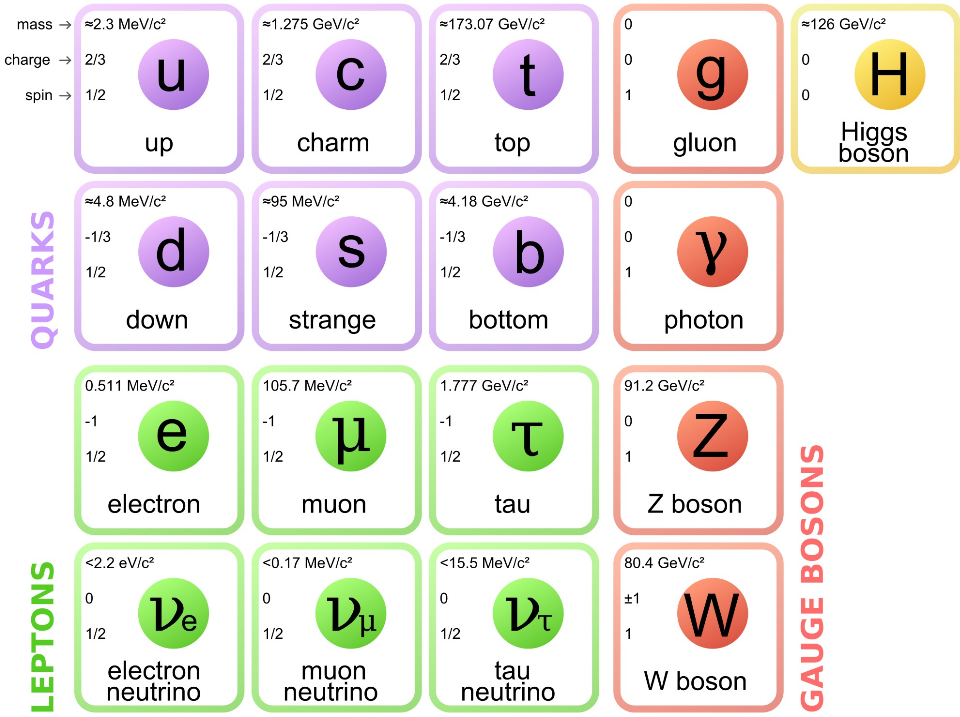


***Figure 2.1:** Particles in the SM*

of angular momentum the sum of the spin (S_e) of the electron, the spin of the resulting nucleus (S_{N^*}) and the angular momentum between them (L) should equal the original nucleus' spin (S_N). Despite this, it was know experimentally that the resulting electrons where emitted in a spectrum of energies less than expected and at $L = 0$, making the sum of spins impossible to match. Pauli's solution to the apparent violation of the conservation of these quantities was to hypothesize the existence of a second neutral spin-$\frac{1}{2}$ with very small mass compared to the electron that allows for both the distribution of energy between both emitted particle, thus letting the electron be emitted at different energies depending on the kinematics, and the conservation of angular momentum.

$$\begin{array}{ccc} N \longrightarrow N^* + e^- & \longrightarrow & N \longrightarrow N^* + e^- + \nu \\ \hline m_N = m_{N^*} + E_{e^-} & \longrightarrow & m_N = m_{N^*} + E_{e^-} + E_\nu \\ S_{z\ N} = S_{z\ N^*} + S_{z\ e^-} & \longrightarrow & S_{z\ N} = S_{z\ N^*} + S_{z\ e^-} + S_{z\ \nu} \end{array}$$

Later refinement of the atomic model found that nuclei where made up of positively charged elements named protons (p) and neutrally charged elements known as neutrons[1] (n), leaving the small hypothesized particle to be named neutrino (ν) later by Enrico Fermi. It is well understood now that the beta decay is a process in which a neutron within an unstable nucleus decays into a proton, an electron and a(n) (anti)neutrino.

$$n \longrightarrow p + e^- + \bar{\nu}$$

Subsequent experiments exploiting cross symmetry of the above interaction [2], accelerator-produced pion decay [3], and charged-current tau neutrino interactions in nuclear emulsion [4, 5] confirmed the existence of three distinct active neutrino

[1] this was the name first proposed for the particle hypothesized by Pauli, later changed

flavors and the conservation of lepton family number in the interactions that involved these particles. Given that neutrinos don't interact via electromagnetic or strong forces a theory of a different force that mediated these interactions was developed into what is now known as the "weak nuclear force".

2.2.2 Weak Nuclear Force

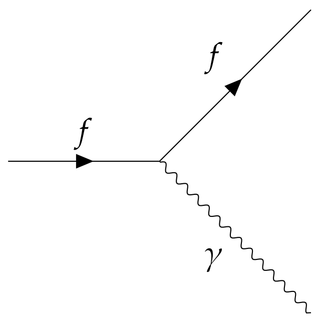
Developments in quantum mechanics and special relativity led up to the first theories that could describe the behavior of some subatomic particles and their interactions through excitations of quantized fields called quantum field theories (QFTs) obtained by promoting classical fields to operator-valued quantities satisfying canonical (anti)commutation relations and deriving their dynamics from a Lorentz-invariant action principle. One of the earliest and most successful QFTs was Quantum Electro Dynamics (QED), describing how spin-1/2 (anti)particles ($\overset{(-)}{\psi}$) interacted with an EM 4-potential (A^{μ}) by minimizing the action given by the Lagrangian density:

$$\mathcal{L}_{\text{QED}} = \underbrace{\bar{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi}_{\text{free Dirac field}} - \underbrace{\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{free EM field}} - \underbrace{e\bar{\psi}\gamma^{\mu}\psi A_{\mu}}_{\text{interaction}}. \tag{2.1}$$

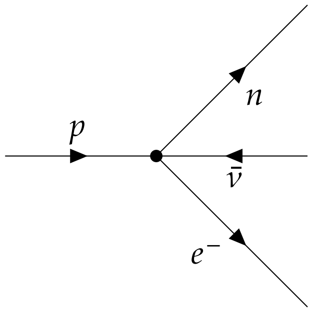
where

$$F_{\mu\nu} = \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu} \tag{2.2}$$

is the electromagnetic field strength tensor, ψ is the charged fermion field of mass m, and $D_{\mu} = \partial_{\mu} + ieA_{\mu}$ is the covariant derivative encoding the minimal coupling between the fermion and the gauge field A_{μ}. While the 'free' terms in (2.1) describe the propagation of Dirac fermions and photons in free Minkowski space, the 'interaction' term describes a vertex that couples the (anti)fermion current with the electromagnetic four-potential, representing the elementary process of a(n) (anti)fermion emitting or absorbing a photon as ilustrated in (2.3) as a Feynman diagram. At zeroth-order $\hbar$ perturbative terms, QED had shown predictive power for processes like Compton ($\gamma + e^{-} \rightarrow \gamma + e^{-}$) [6] and Møller ($e^{-} + e^{-} \rightarrow e^{-} + e^{-}$) [7] scattering by the time Fermi first proposed an interaction inspired by the QED interaction term, where instead of a fermion current emits a photon, two fermion currents meet at a vertex such that a coupling constant (now called Fermi's constant G_F) quantified the intensity of the interaction, and thus can be estimated experimentally. This new interaction vertex is shown in (2.4) as an effective Feynman diagram.



$$\sim (\overline{\psi}_f e Q_f \gamma^{\mu}\psi_f)A_{\mu} \tag{2.3}$$



$$\sim \frac{G_F}{\sqrt{2}}(\overline{\psi}_n \gamma^\mu \psi_p)(\overline{\psi}_\nu \gamma_\mu \psi_{e^-}) \tag{2.4}$$

Subsequent discoveries in particle physics revealed that charged-current weak interactions violate parity symmetry maximally. First theorized by Lee and Yang in 1956 [8] to resolve the then-known $\tau - \theta$ puzzle, and later verified experimentally by Wu et al. in 1957 [9], this motivated a modification of the Fermi interaction in the form of a combination of the two bilinear Lorentz-invariant structures that admit such violation [10, 11].

$$\text{V} : j_V^\mu = \bar{\psi}\gamma^\mu\psi \qquad \xrightarrow{\mathcal{P}} \qquad j'_{V\,\mu} = \bar{\psi}\gamma_\mu\psi \tag{2.5}$$

$$\text{A} : j_A^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi \qquad \xrightarrow{\mathcal{P}} \qquad j'_{A\,\mu} = -\bar{\psi}\gamma_\mu\gamma^5\psi \tag{2.6}$$

The expressions (2.5) and (2.6) show how different bilinear structures change under parity ($\mathcal{P}$) and how pure vector (axial) currents are $\mathcal{P} - even$ ($\mathcal{P} - odd$). Given a general current structure, $j^\mu = g_V j_V^\mu + g_A j_A^\mu$, observables that depend on the inner product of two such currents transform under parity as:

$$|\mathcal{M}|^2 = (j_1 \cdot j_2) = g_V^2(j_{1\,V} \cdot j_{2\,V}) + g_A^2(j_{1\,A} \cdot j_{2\,A}) + g_V g_A(j_{1\,V} \cdot j_{2\,A} + j_{1\,A} \cdot j_{2\,V}) \tag{2.7}$$

$$\xrightarrow{\mathcal{P}}$$

$$|\mathcal{M}'|^2 = (j'_1 \cdot j'_2) = g_V^2(j'_{1\,V} \cdot j'_{2\,V}) + g_A^2(j'_{1\,A} \cdot j'_{2\,A}) - g_V g_A(j'_{1\,V} \cdot j'_{2\,A} + j'_{1\,A} \cdot j'_{2\,V}). \tag{2.8}$$

The only difference between (2.7) and (2.8) lies in the sign of the interference term, such that the degree of $\mathcal{P}$-symmetry violation is given by the coefficient $\frac{g_V g_A}{g_V^2 + g_A^2}$. Normalizing this there are two options for maximally violated $\mathcal{P}$-symmetry given by $\frac{g_V}{g_A} = \pm 1$, also known as $V \pm A$ currents. Given that in the Wu experiment it was determined that (anti)neutrinos are emitted exclusively as left(right)-handed fermions the appropriate structure to use for weak interactions was determined to be $V - A$. Modifying the Fermi Lagrangian density for beta decay to include this gives the following expression:

$$\mathcal{L}_{CC} = \frac{G_F}{\sqrt{2}} \left[\bar{\psi}_{\nu_e}\gamma^\mu(1-\gamma^5)\psi_e\right] \left[\bar{\psi}_p\gamma_\mu(1-\gamma^5)\psi_n\right] + \text{h.c.} \tag{2.9}$$

Later refinements of this theory succeeded in demonstrating the universality of the weak interaction across leptonic decays and strangeness-changing meson decays through Cabibbo mixing [12]. This not only established that all such processes are mediated by the same underlying force, but also motivated the prediction and subsequent discovery of the charm quark [13]. With these improvements on the theory thus far, the point-like structure of the interaction still led to scattering amplitudes proportional to the center-of-mass energy $\sqrt{s}$, resulting in S-matrix unitarity violation at higher energies. The natural resolution, in analogy with QED, is to conceive of Fermi's interaction as the low-energy limit of a gauge field theory, wherein the maximal parity violation of the charged currents naturally gives rise to three mediating vector bosons corresponding to the generators of an $SU(2)_L$ symmetry, which must be massive due to the short range of the weak interaction. Given this hypothesis, Glashow showed in 1961 [14] that the gauge field theory whose low-energy limit reproduces the $V - A$ structure of charged currents requires a $SU(2)_L \times U(1)_Y$ symmetry group, which not only predicted the existence of neutral weak currents, later measured in 1973 at the Gargamelle Neutrino Experiment at CERN [15, 16], but also unified the weak and electromagnetic interactions into a single electroweak theory, later completed independently by Weinberg and Salam through the incorporation of spontaneous symmetry breaking [17, 18].

2.3. Massive and Massless Mediators

This section describes the implementation of different types of gauge bosons as mediators of the relevant forces involved in the interactions of this work. These include the photon as a massless spin-1 mediator of the EM force and the $W^{\pm}$, Z^0 bosons as massive spin-1 mediators of the weak nuclear force.

2.3.1 Photons

It is well established within QED that electromagnetic interactions between charged fermions are mediated by a bosonic gauge field. The mediating boson, the photon, is massless as required by $U(1)$ gauge invariance, a property that manifests physically as the infinite range of the electromagnetic interaction, and carries spin-1. Its dynamics are governed by (2.1), and by applying the Euler-Lagrange equations to the free electromagnetic field term with respect to A^μ, one obtains the equations of motion:

$$\partial_\mu \left(\frac{\partial \mathcal{L}_{\text{free EM}}}{\partial(\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}_{\text{free EM}}}{\partial A_\nu} = 0 \implies \partial_\mu F^{\mu\nu} = \partial^2 A^\nu - \partial^\nu (\partial_\mu A^\mu) = 0, \tag{2.10}$$

which in momentum space, with $\partial_\mu \to -iq_\mu$, becomes

$$\left(-q^2 \eta^{\mu\nu} + q^\mu q^\nu\right) \widetilde{A}_\nu = K^{\mu\nu} \widetilde{A}_\nu = 0. \tag{2.11}$$

The propagator $D^{\mu\nu}$, defined by $K^{\mu\alpha} D_{\alpha\nu} = \delta^\mu{}_\nu$, cannot be obtained directly from the kinetic operator $K^{\mu\nu}$, since the latter is singular by construction: the action is invariant under $U(1)$ gauge transformations $A^\mu \to A'^\mu = A^\mu + \partial^\mu \lambda$, which implies that $K^{\mu\nu}$ has a zero eigenvalue along the gauge orbit. A propagator is only obtainable upon gauge fixing, which is implemented by augmenting the Lagrangian density with a gauge-fixing term,

$$\mathcal{L}'_{\text{free EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} \left(\partial_\mu A^\mu\right)^2, \tag{2.12}$$

such that the modified kinetic operator

$$K'^{\mu\nu} = -q^2 \eta^{\mu\nu} + \left(1 - \frac{1}{\xi}\right) q^\mu q^\nu \tag{2.13}$$

is now invertible for any $\xi \neq 0$, yielding the generalized R_ξ propagator

$$D_\gamma^{\mu\nu}(q) = \frac{-i}{q^2 + i\epsilon}\left[\eta^{\mu\nu} - (1-\xi)\frac{q^\mu q^\nu}{q^2}\right], \tag{2.14}$$

where any choice of $\xi \in \mathbb{R} \setminus \{0\}$ yields equivalent physical predictions, as gauge-dependent terms cancel in all observable quantities.

2.3.2 $W^\pm$ and Z

In contrast to the photon, the weak interaction's short range is a consequence of its mediators' mass and thus its dynamics are no longer governed by a Maxwell Lagrangian density, but rather one that includes a mass term that breaks the $U(1)$ found in the original, called a Proca Lagrangian, with a propagator that no longer requires gauge fixing but still is invariant under Lorentz transformations.

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2 A_\mu A^\mu. \tag{2.15}$$

In calculating its propagator by solving the equations of motion and inverting the kinetic operator in momentum space one finds that:

$$\partial_\mu\left(\frac{\partial\mathcal{L}_{\text{Proca}}}{\partial(\partial_\mu A_\nu)}\right) - \frac{\partial\mathcal{L}_{\text{Proca}}}{\partial A_\nu} = 0 \implies \partial_\mu F^{\mu\nu} + M^2 A^\nu = \partial^2 A^\nu - \partial^\nu(\partial_\mu A^\mu) + M^2 A^\nu = 0, \tag{2.16}$$

and by taking the four-divergence ∂_ν of (2.16),

$$\begin{aligned} \partial_\nu\partial^2 A^\nu - \partial_\nu\partial^\nu(\partial_\mu A^\mu) + M^2(\partial_\mu A^\nu) &= 0, \\ \implies M^2(\partial_\mu A^\nu) &= 0. \end{aligned} \tag{2.17}$$

Now taking $M^2 \neq 0$ the gauge given by $\partial_\mu A^\mu = 0$ is no longer a choice but rather an equation of motion that reduces the four degrees of freedom of the components of A^μ to two transverse and one longitudinal polarization modes of a massive spin-1 particle. Writing the equations of motion in momentum space with $\partial^\mu \to -iq^\mu$:

$$(-q^2\eta^{\mu\nu} + q^\nu q_\mu + M^2)A_\mu = ((M^2 - q^2)\eta^{\mu\nu} + q^\nu q_\mu)A_\mu = 0. \tag{2.18}$$

Inverting $K^{\mu\nu}_{\text{Proca}} = ((M^2 - q^2)\eta^{\mu\nu} + q^\mu q^\nu)$ gives a unique choice for the Proca propagator $D^{\mu\nu}_{\text{Proca}}$ such that $K^{\mu\alpha}_{\text{Proca}} D_{\text{Proca}\,\alpha\nu} = \delta^\mu{}_\nu$.

$$D^{\mu\nu}_{\text{Proca}} = \frac{i}{q^2 - M^2 + i\epsilon}\left[\eta^{\mu\nu} - \frac{q^\mu q^\nu}{M^2}\right]. \tag{2.19}$$

It is worth noting that interactions mediated by (2.19) between conserved currents, for which $\partial_\mu j^\mu = 0$ implies $q_\mu \widetilde{j}^\mu = 0$, reduce to the Fermi effective form in the limit $|q^2| \ll M^2$. Expanding the denominator,

$$\frac{1}{q^2 - M^2} = \frac{-1}{M^2}\left[\frac{1}{1 - q^2/M^2}\right] \xrightarrow{|q^2| \ll M^2} \frac{-1}{M^2}\left[1 + \frac{q^2}{M^2} + \mathcal{O}\left(\frac{q^4}{M^4}\right)\right], \tag{2.20}$$

where the amplitude of an interaction is now

$$i\mathcal{M} = (-ig)^2\, j_\mu^\dagger\, D^{\mu\nu}_{\text{Proca}}\, j_\nu \xrightarrow{|q^2|\ll M^2} \frac{ig^2}{M^2}\, j_\mu^\dagger\, \eta^{\mu\nu}\, j_\nu, \tag{2.21}$$

which upon comparison with the Fermi four-fermion amplitude $i\mathcal{M}_{\text{Fermi}} = i\frac{G_F}{\sqrt{2}}\, j_\mu^\dagger\, \eta^{\mu\nu}\, j_\nu$ yields

$$\frac{G_F}{\sqrt{2}} \sim \frac{g^2}{M^2}, \tag{2.22}$$

where its actual matching relation will be given when calculating the $SU(2)_L$ couplings with $W^\pm$ and charged currents.

2.4. Electroweak mixing and $\sin^2\theta_W$

This section describes how the electromagnetic and weak force can be described as a single unified force and how its spontaneous symmetry breaking provides the origin of mass-terms for the $W^\pm$ and Z^0 gauge bosons.

2.4.1 $SU(2)_L$

As established in the previous section, the inclusion of massive spin-1 bosons explicitly breaks the usual $U(1)$ symmetry found in QED and thus a gauge theory that accommodates such mediators requires a larger symmetry group. A natural candidate comes in the study of a theory with an $SU(2)$ symmetry whose three generators,

$$\tau^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \tau^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \tau^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{2.23}$$

couple exclusively with left(right)-handed (anti)fermions (as is the case for charged current weak interactions) that act on doublets of the form

$$L = \begin{pmatrix} \nu_l \\ l \end{pmatrix}_L, \tag{2.24}$$

where l denotes any charged lepton and ν_l it's corresponding (anti)neutrino. A theory of fermion interactions with a local $SU(2)_L$ symmetry requires promoting the partial derivative in the Dirac Lagrangian to a covariant derivative

$$D_\mu = \partial_\mu + ig\frac{\tau^a}{2}W^a_\mu = \partial_\mu + ig\left(\frac{\tau^1}{2}W^1_\mu + \frac{\tau^2}{2}W^2_\mu + \frac{\tau^3}{2}W^3_\mu\right), \tag{2.25}$$

and introducing a non-abelian field strength tensor for each gauge boson W^a, $a \in \{1,2,3\}$,

$$W^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu - g\epsilon^{abc}W^b_\mu W^c_\nu, \tag{2.26}$$

where the last term, absent in the abelian case of QED, encodes the self-interactions among the gauge bosons that arise from the non-abelian nature of $SU(2)_L$. The resulting Lagrangian density is

$$\mathcal{L}_{SU(2)_L} = \bar{L}\, i\gamma^\mu D_\mu L - \frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu}, \tag{2.27}$$

where $L = \binom{\nu_l}{l}_L$ denotes the left-handed lepton doublet. Extracting and expanding the interaction term in (2.27) explicitly gives,

$$\begin{aligned}\mathcal{L}_{SU(2)_L,\,\mathrm{int}} &= -\frac{g}{2}\bar{L}\gamma^\mu(\tau^1 W^1_\mu + \tau^2 W^2_\mu + \tau^3 W^3_\mu)L \\ &= -\frac{g}{2}\begin{pmatrix}\bar{\nu}_l & \bar{l}\end{pmatrix}_L \gamma^\mu \begin{pmatrix} W^3_\mu & W^1_\mu - iW^2_\mu \\ W^1_\mu + iW^2_\mu & -W^3_\mu\end{pmatrix}\begin{pmatrix}\nu_l \\ l\end{pmatrix}_L .\end{aligned} \tag{2.28}$$

Identifying that the upper right and lower left entries of the matrix in (2.28) correspond neatly to the known charged current weak interactions makes the definition of the $W^\pm$ bosons such that $W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp iW^2_\mu)$ a way to separate the interaction Lagrangian density into a charged current (CC) term and a neutral current (NC) term.

$$\mathcal{L}_{\mathrm{int}} = \underbrace{-\frac{g}{2}\left[\bar{\nu}_L\gamma^\mu\nu_L - \bar{l}_L\gamma^\mu l_L\right]W^3_\mu}_{\mathcal{L}_{\mathrm{NC\,int}}}\underbrace{-\frac{g}{\sqrt{2}}\left[\bar{\nu}_L\gamma^\mu l_L\, W^+_\mu + \bar{l}_L\gamma^\mu\nu_L\, W^-_\mu\right]}_{\mathcal{L}_{\mathrm{CC\,int}}}. \tag{2.29}$$

It is worth noting that the amplitude of a CC interaction computed from $\mathcal{L}_{CC\,\mathrm{int}}$ in (2.29) in the limit $|q^2| \ll M_W^2$ reproduces the Fermi four-fermion amplitude, yielding the tree-level matching relation between G_F and g.

$$\begin{aligned} -i\mathcal{M} &= \left[\frac{-ig}{\sqrt{2}}\bar{\nu}_L\gamma_\mu l_L\right]\frac{i\eta^{\mu\nu}}{M_W^2 - q^2}\left[\frac{-ig}{\sqrt{2}}\bar{l}_L\gamma_\nu\nu_L\right] \qquad (2.30)\\ &= \left[\frac{-ig}{\sqrt{2}}\bar{\nu}\frac{1}{2}\gamma_\mu(1-\gamma^5)l\right]\frac{i\eta^{\mu\nu}}{M_W^2 - q^2}\left[\frac{-ig}{\sqrt{2}}\bar{l}\gamma_\nu(1-\gamma^5)\nu\right] \\ &\overset{|q^2|\ll M_W^2}{\approx} -\frac{ig^2}{8M_W^2}\left[\bar{\nu}\gamma^\mu(1-\gamma^5)l\right]\left[\bar{l}\gamma^\mu(1-\gamma^5)\nu\right]. \qquad (2.31)\end{aligned}$$

Comparing (2.31) with the amplitude given by (2.9) yields the relation

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}, \tag{2.32}$$

which is of particular importance as it expresses the experimentally well-measured Fermi constant directly in terms of the $SU(2)_L$ gauge coupling and the $W^\pm$ mass, thereby connecting the low-energy effective theory to the underlying electroweak gauge structure and providing a precise prediction for M_W. Given this and the knowledge that weak neutral currents exist, it is tempting to identify W^3_μ as their mediator. However, this cannot be the case since this term couples exclusively to left(right)-handed (anti)fermions, while it is experimentally well established that weak neutral current interactions do not maximally violate $\mathcal{P}$-symmetry as charged current interactions do [15, 16]. The $SU(2)_L$ symmetry is therefore too chiral to accommodate weak neutral currents, which while not $\mathcal{P}$-conserving, do not exhibit the maximal parity violation of the charged current sector. This incompatibility however points to the possibility of a more complete theory wherein the purely chiral $SU(2)_L$ generators are combined with a $\mathcal{P}$-conserving $U(1)_Y$ hypercharge gauge field, analogous to that of QED, to model the weak and electromagnetic interactions as a single unified force known as the Electroweak (EW) theory.

Multiplet	T^3	Q	Y
$\nu_{l\,L}$	$\frac{1}{2}$	0	-1
l_L	-$\frac{1}{2}$	-1	-1
l_R	0	-1	-2

Table 2.1: *Values for Q, T^3 and Y for each lepton, where l denotes any of the known flavours of charged lepton*

2.4.2 $SU(2)_L \times U(1)_Y$

To establish a theory coherent with the known properties of leptons, it is first necessary to fix the hypercharge (Y) assignments of the left-handed fermion doublet $L = \binom{\nu_l}{l}_L$ and the right-handed lepton singlet $R = (l_R)$ by relating them to the known electric charges Q and weak isospin eigenvalues T^3 via the Gell-Mann–Nishijima relation

$$Q = T^3 + \frac{Y}{2}. \tag{2.33}$$

Fixing these gives the values presented in Table 2.1 for each lepton type, noting that right-handed neutrinos are absent from the spectrum due to the lack of experimental evidence for their existence, with the exception of the measured neutrino mass splittings among the three flavors, which falls outside the scope of this work. With this, the full Lagrangian density with the necessary assigned values is:

$$\mathcal{L}_{EW} = -i\bar{L}\gamma^\mu D_\mu L - i\bar{R}\gamma^\mu D_\mu R - \frac{1}{4}W^{a\,\mu\nu}W^a_{\mu\nu} - \frac{1}{4}B^{\mu\nu}B_{\mu\nu} \tag{2.34}$$

Where $B^{\mu\nu}$ is defined the same way as $F^{\mu\nu}$ in (2.2) for a massless boson field B^μ. The covariant derivative is given by

$$D_\mu = \partial_\mu + ig(\frac{\tau^1}{2}W^1_\mu + \frac{\tau^2}{2}W^2_\mu + \frac{\tau^3}{2}W^3_\mu) + ig'\frac{Y}{2}B_\mu. \tag{2.35}$$

Much like (2.28), the interaction terms of (2.34) can be separated into CC and NC contributions, wherein the NC terms can be re-expressed as a linear combination of two mass eigenstates mixed through an orthogonal rotation parametrized by a single angle known as the Weinberg angle (θ_W). The EW interaction terms are given by

$$\begin{aligned}\mathcal{L}_{EW\ \mathrm{int}} &= -\frac{g}{2}\begin{pmatrix}\bar{\nu}_l & \bar{l}\end{pmatrix}_L \gamma^\mu \begin{pmatrix} W^3_\mu & W^1_\mu - iW^2_\mu \\ W^1_\mu + iW^2_\mu & -W^3_\mu\end{pmatrix}\begin{pmatrix}\nu_l \\ l\end{pmatrix}_L - \frac{g'}{2}\left(\bar{\psi}\gamma^\mu Y\psi\right)B_\mu \\ &= \underbrace{-\frac{g}{\sqrt{2}}\left[\bar{\nu}_L\gamma^\mu l_L\, W^+_\mu + \bar{l}_L\gamma^\mu\nu_L\, W^-_\mu\right]}_{\mathcal{L}_{CC\,\mathrm{int}}}\underbrace{-\frac{g}{2}\left[\bar{\nu}_L\gamma^\mu\nu_L - \bar{l}_L\gamma^\mu l_L\right]W^3_\mu - \frac{g'}{2}\bar{\psi}\gamma^\mu Y B_\mu\psi}_{\mathcal{L}_{NC\,\mathrm{int}}},\end{aligned} \tag{2.36}$$

where ψ denotes fermions with hypercharge Y and weak isospin T^3, the latter being explicitly dependent on the chirality of the field, as left-handed fermions carry $T^3 =$

$\pm 1/2$ while right-handed fermions are $SU(2)_L$ singlets with $T^3 = 0$.

$$\begin{aligned}\mathcal{L}_{NC\,\text{int}} &= -\frac{g}{2}\left[\bar{\nu}_L\gamma^\mu\nu_L - \bar{l}_L\gamma^\mu l_L\right]W^3_\mu - \frac{g'}{2}\bar{\psi}\gamma^\mu Y B_\mu\psi \\ &= -\frac{g}{2}\bar{\psi}\gamma^\mu(2T^3)\psi - \frac{g'}{2}\bar{\psi}\gamma^\mu Y B_\mu\psi \\ &= -\bar{\psi}\gamma^\mu(gT^3W^3_\mu + g'\frac{Y}{2}B_\mu)\psi.\end{aligned} \tag{2.37}$$

The W^3_μ and B_μ fields correspond to the unmixed states that can be combined into the known mass eigenstates A_μ and Z_μ through the orthogonal transformation given by:

$$\begin{pmatrix} B_\mu \\ W^3_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W \end{pmatrix}\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix}. \tag{2.38}$$

Substituting (2.38) into (2.37),

$$\begin{aligned}\mathcal{L}_{NC\,\text{int}} = &-\left(g\sin\theta_W T^3 + g'\cos\theta_W\frac{Y}{2}\right)\bar{\psi}\gamma^\mu\psi\, A_\mu \\ &-\left(g\cos\theta_W T^3 - g'\sin\theta_W\frac{Y}{2}\right)\bar{\psi}\gamma^\mu\psi\, Z_\mu.\end{aligned} \tag{2.39}$$

Requiring that A_μ corresponds to the photon field that couples with fermions such that $\mathcal{L}_{\text{QED int}} = -e\overline{\psi}\gamma^\mu Q\psi$ where Q follows from (2.33) gives the relations between the coupling constants

$$e = g\sin\theta_W = g'\cos\theta_W. \tag{2.40}$$

Substituting (2.33) and (2.40) into the Z_μ coefficient of (2.39) yields the Z boson interaction term expressed solely in terms of T^3 and Q,

$$\mathcal{L}_{NC\,\text{int}} = -eQ\bar{\psi}\gamma^\mu\psi\, A_\mu - \frac{g}{\cos\theta_W}\left(T^3 - Q\sin^2\theta_W\right)\bar{\psi}\gamma^\mu\psi\, Z_\mu. \tag{2.41}$$

It will be useful to later sections of this work to have the Z boson interaction term separated into vector and axial-vector terms such that for a fermion f,

$$\mathcal{L}_{Zf\bar{f}\,\text{int}} = -\frac{g}{2\cos\theta_W}\bar{\psi}_f\gamma^\mu(g^f_V - g^f_A\gamma^5)\psi_f Z_\mu. \tag{2.42}$$

The vector (g^f_V) and axial-vector (g^f_A) coupling coefficients can be calculated explicitly noting that all right-handed fermions are $SU(2)_L$ singlets and thus $T^3P_R\psi = T^3\frac{1}{2}(1+\gamma^5)\psi = T^3\psi_R = 0$. Separating the Z interaction term in (2.41),

$$\begin{aligned}\mathcal{L}_{Zf\bar{f}\,\text{int}} &= -\frac{g}{\cos\theta_W}\left(T^3_f - Q_f\sin^2\theta_W\right)\bar{\psi}_f\gamma^\mu\psi_f\, Z_\mu \\ &= -\frac{g}{\cos\theta_W}\bar{\psi}_f\frac{\gamma^\mu}{2}\left(T^3_f - Q_f\sin^2\theta_W\right)(1 + 1 + \gamma^5 - \gamma^5)\psi_f\, Z_\mu \\ &= -\frac{g}{\cos\theta_W}\bar{\psi}_f\frac{\gamma^\mu}{2}\left(\cancelto{0}{T^3_f} - Q_f\sin^2\theta_W\right)(1 + \gamma^5)\psi_f\, Z_\mu\end{aligned}$$

$$
\begin{aligned}
&-\frac{g}{\cos\theta_W}\bar{\psi}_f\frac{\gamma^\mu}{2}\left(T_f^3-Q_f\sin^2\theta_W\right)(1-\gamma^5)\psi_f\,Z_\mu \\
&=-\frac{g}{\cos\theta_W}\bar{\psi}_f\gamma^\mu\left((\frac{T_f^3}{2}-Q_f\sin^2\theta_W)-\frac{T^3}{2}\gamma^5\right)\psi_f\,Z_\mu.
\end{aligned}
\tag{2.43}
$$

Comparing (2.43) with (2.42),

$$
\begin{cases}
g_V^f = T_f^3 - 2Q_f\sin^2\theta_W, & (2.44)\\
g_A^f = T_f^3. & (2.45)
\end{cases}
$$

It is worth noting that the presence of the $2Q_f\sin^2\theta_W$ term in g_V^f (2.44) is precisely what prevents the Z boson interaction from being maximally $\mathcal{P}$-violating and thus neutral particles maintain the chiral structure of CC interactions even in NC vertices.

While the $SU(2)_L\times U(1)_Y$ Lagrangian density correctly describes the interaction structure of the $W^\pm$ and Z bosons with fermions, explicit mass terms for these mediators of the form $\frac{1}{2}M^2A_\mu A^\mu$ would break the gauge symmetry of the theory, as shown in (2.15). The resolution, proposed independently by Higgs, Englert and Brout in 1964 [**PhysRevLett.13.585**, 19, 20], is to introduce a complex scalar $SU(2)_L$ doublet ϕ with $U(1)_Y$ hypercharge $Y = 1$, known as the Higgs doublet, whose potential is chosen such that it acquires a non-zero vacuum expectation value $\langle\phi\rangle = \begin{pmatrix}0\\ v/\sqrt{2}\end{pmatrix}$. Expanding the theory around this vacuum spontaneously breaks $SU(2)_L\times U(1)_Y$ down to $U(1)_{\text{EM}}$, leaving the photon massless while generating masses for the $W^\pm$ and Z bosons through the covariant kinetic term $|D_\mu\phi|^2$.

$$
|D_\mu\phi|^2 \supset \frac{v^2}{8}\left[g^2(W_\mu^1)^2+g^2(W_\mu^2)^2+\left(gW_\mu^3-g'B_\mu\right)^2\right],
\tag{2.46}
$$

from which the $W^\pm$ mass is immediately read off as

$$
M_W = \frac{gv}{2},
\tag{2.47}
$$

while the neutral sector mass matrix in the (W_μ^3, B_μ) basis,

$$
\frac{v^2}{8}\begin{pmatrix}W_\mu^3 & B_\mu\end{pmatrix}\begin{pmatrix}g^2 & -gg'\\ -gg' & g'^2\end{pmatrix}\begin{pmatrix}W^{3\mu}\\ B^\mu\end{pmatrix},
\tag{2.48}
$$

has eigenvalues $\lambda_1 = g^2+g'^2$ and $\lambda_2 = 0$, corresponding to the massive Z boson and the massless photon respectively, giving

$$
M_Z = \frac{v}{2}\sqrt{g^2+g'^2}.
\tag{2.49}
$$

Taking the ratio of (2.47) and (2.49) and identifying $\cos\theta_W = g/\sqrt{g^2+g'^2}$ from (2.40) yields the tree-level mass relation

$$
M_W = M_Z\cos\theta_W,
\tag{2.50}
$$

which is one of the most precisely tested predictions of the electroweak theory, with radiative corrections to this relation providing stringent constraints on physics beyond the Standard Model.

The couplings g_V^f and g_A^f also constitute tree-level SM predictions for the vector and axial-vector couplings of neutral current weak interactions, whose values are fixed entirely by T^3, Q, and $\sin^2\theta_W$ for each fermion. However, current experimental precision in measuring observables of such interactions, most notably in neutrino experiments, has not only confirmed these tree-level predictions but also demands that higher-order perturbative corrections in the form of loop diagrams be taken into account. These radiative corrections introduce a shift in the value of g_V^f (or more precisely $\sin^2\theta_W$) as will be discussed in Section 3.5 such that the new value differs measurably between the Z pole and the low-energy regime probed by reactor neutrino experiments, and acquire physical meaning when interpreted as corrections to the $ff\gamma$ vertex function. The following chapter focuses on how, despite carrying no electric charge, the neutrino nonetheless possesses measurable electromagnetic properties predicted by the SM through precisely these radiative corrections.

3. Radiative Corrections and Neutrino Charge Radius

This chapter is concerned with the main theoretical focus of this work. The Standard Model (SM) as described in Chapter 2, and in specific Electroweak (EW) theory within it, models the interactions between fermions and gauge bosons utilizing of perturbative terms in the from of higher-order loop diagrams. While tree level calculations offered good agreement with experimental data, modern precision in particle experiments have required in recent history have shown the necessity to include higher-order terms to either validate the SM or in occasions point to scenarios Beyond the SM (BSM). This chapter describes how the inclusion of 1-loop diagrams to an effective interaction reveal predictions of electromagnetic properties of the neutrino and how they modify previously established parameters in the SM.

3.1. Introduction

The electromagnetic properties of the neutrino were not only essential to its first postulation and subsequent discovery, but remain an active field of research in modern experimental particle physics. Indeed, both its neutral charge and the possibility of a non-zero magnetic dipole moment are mentioned in the original 1930 letter by Pauli that first introduced the neutrino [1]. The study of such properties was further motivated by the proposal and eventual discovery of neutrino flavor oscillations and their requisite non-zero mass splittings. This chapter describes how radiative corrections to the effective $\nu\nu\gamma$ vertex predict non-trivial electromagnetic form factors for massive[1] neutrinos within the SM, the focus of this work being the neutrino charge radius (NCR).

3.2. Electromagnetic Form Factors

This section presents the derivation of electromagnetic properties from effective neutrino–photon interactions for massive Dirac neutrinos and ultra-relativistic (practically massless) Weyl neutrinos.

3.2.1 For Massive Neutrinos

Following Ref. [23], the interaction between a fermionic field $\mathfrak{f}(x)$ and the electromagnetic field A^μ in the SM is given by the interaction Hamiltonian

$$\mathcal{H}_{\mathrm{EM}}(x) = j^{\mathfrak{f}}_\mu(x) A^\mu(x) = q_{\mathfrak{f}} \bar{\mathfrak{f}}(x)\gamma_\mu \mathfrak{f}(x) A^\mu(x), \tag{3.1}$$

where $q_{\mathfrak{f}}$ is the electric charge of a fermion $\mathfrak{f}$. Fig. 3.1(a) shows the corresponding the tree-level Feynman diagram for (3.1) where the photon γ is the quanta of the EM field

[1]This chapter concerns primarily Dirac-type neutrinos; analogous calculations for Majorana neutrinos can be found in [21, 22, 23]

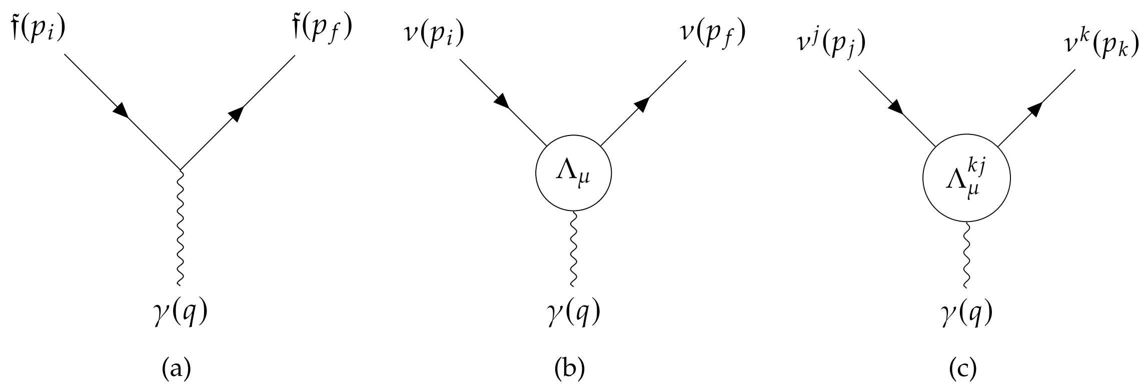


Figure 3.1: *Charged fermion–photon and effective neutrino–photon interaction vertices.*

A^μ. For the neutrino no such direct interaction term exists due to its neutral charge. Nevertheless, at the quantum level such interactions can arise when including higher order terms in the perturbative expansion of interaction terms in the form of loop diagrams. In the 1-photon approximation the effective interaction Hamiltonian for a massive neutrino ν is given by

$$\mathcal{H}_{\mathrm{EM}}^{(\nu)}(x) = j_\mu^{(\nu)}(x)A^\mu(x) = \bar{\nu}(x)\Lambda_\mu \nu(x)A^\mu(x), \tag{3.2}$$

where $j_\mu^{(\nu)}(x)$ is the effective neutrino current and Λ_μ is a 4×4 vertex function matrix in spinor space which can contain space-time derivatives such that $j_\mu^{(\nu)}(x)$ transforms as a four-vector. The corresponding effective Feynman diagram for (3.2) is shown in Fig. 3.1(b). We will see that the neutrino electromagnetic form factors and properties such as charge and magnetic dipole will correspond to the Fig. 3.1(b) interaction, but it is worth contrasting the origin of these form factors with the analogous quantities for composite particles such as the proton or the neutron, where electromagnetic form factors encode genuine information about the spatial distribution of fundamental constituents. The neutrino carries no such substructure; its form factors are instead radiative effects generated by quantum loops involving gauge bosons. They are physical and, in principle, measurable, but they constitute effective electromagnetic properties induced by the weak interaction rather than a reflection of any internal composition. In this sense the neutrino electromagnetic form factors occupy a conceptually distinct place within the Standard Model.
We are intrested in the neutrino part of the Fig. 3.1(b) interaction amplitude, given by the matrix element

$$\langle \nu(p_f, h_f)|\, j_\mu^{(\nu)}(x)\, |\nu(p_i, h_i)\rangle\,, \tag{3.3}$$

where $p_{i(f)}$ and $h_{i(f)}$ denote the momentum and helicity of the initial (final) neutrino respectively. Setting aside the helicity dependence for now and taking into account

$$\partial^\mu j_\mu^{(\nu)} = i\left[\mathcal{P}^\mu, j_\mu^{(\nu)}\right], \tag{3.4}$$

where $\mathcal{P}^\mu$ is the four-momentum operator which generates translations, the effective neutrino current can be rewritten as

$$j_\mu^{(\nu)}(x) = e^{i\mathcal{P}\cdot x} j_\mu^{(\nu)}(0) e^{-i\mathcal{P}\cdot x}. \tag{3.5}$$

Since $\mathcal{P}^\mu\,|\nu(p_{i(f)})\rangle = p^\mu_{i(f)}\,|\nu(p_{i(f)})\rangle$ substituting (3.5) into (3.3) gives

$$\langle \nu(p_f)|\, j^{(\nu)}_\mu(x)\,|\nu(p_i)\rangle = e^{(p_f-p_i)\cdot x}\,\langle \nu(p_f)|\, j^{(\nu)}_\mu(0)\,|\nu(p_i)\rangle\,. \tag{3.6}$$

It is evident from (3.6) that the quantity encoding the neutrino–photon interaction is the matrix element $\langle \nu(p_f)|\, j^{(\nu)}_\mu(0)\,|\nu(p_i)\rangle$. Since $|\nu(p_i)\rangle$ and $\langle \nu(p_f)|$ describe incoming and outgoing free Dirac particles whose field admits the Fourier expansion

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3 2E}\sum_{h=\pm}\left[a^{(h)}(p)\,u^{(h)}(p)\,e^{-ix\cdot p} + b^{(h)\dagger}(p)\,v^{(h)}(p)\,e^{ix\cdot p}\right], \tag{3.7}$$

the only spinor structures that survive in (3.6) are those sandwiched between free-particle spinors, so that

$$\langle \nu(p_f)|\, j^{(\nu)}_\mu(0)\,|\nu(p_i)\rangle = \bar{u}(p_f)\,\Lambda_\mu(p_f,p_i)\,u(p_i). \tag{3.8}$$

Here Λ_μ is a 4×4 matrix in spinor space whose free Lorentz index must transform as a contravariant four-vector. The most general such matrix constructed from γ-matrices and the available four-momenta $p_{i,f}$, or equivalently from $P_\mu = (p_f+p_i)_\mu$ and $q_\mu = (p_f-p_i)_\mu$, is [24]

$$\Lambda_\mu = \mathbb{f}_1(q^2)\,q_\mu + \mathbb{f}_2(q^2)\,q_\mu\gamma_5 + \mathbb{f}_3(q^2)\,\gamma_\mu + \mathbb{f}_4(q^2)\,\gamma_\mu\gamma_5 + \mathbb{f}_5(q^2)\,\sigma_{\mu\nu}q^\nu + \mathbb{f}_6(q^2)\,\epsilon_{\mu\nu\rho\lambda}\,q^\nu\sigma^{\rho\lambda}, \tag{3.9}$$

where $\mathbb{f}_k$ ($k = 1,\cdots,6$) are six Lorentz-invariant form factors and q is the four-momentum transferred to the photon. Its worth noting that q^2 is the only available independent Lorentz-invariant kinematic quantity since $(p_i+p_f)^2 = 4m_\nu^2 - q^2$ and thus the vertex function $\Lambda_\mu(q)$ can be written as only dependent on q. Further restrictions on $\Lambda_\mu(q)$ are seen when taking into account that $\mathcal{H}^{(\nu)}_{\text{EM}} = \mathcal{H}^{(\nu)\dagger}_{\text{EM}}$ and $A_\mu = A^\dagger_\mu$ thus

$$\langle \nu(p_f)|\, j^{(\nu)}_\mu(0)\,|\nu(p_i)\rangle = \langle \nu(p_f)|\, j^{(\nu)}_\mu(0)\,|\nu(p_i)\rangle^\dagger \tag{3.10}$$

$$\implies \bar{u}(p_f)\,\Lambda_\mu(q)\,u(p_i) = \left(\bar{u}(p_f)\,\Lambda_\mu(q)\,u(p_i)\right)^\dagger \tag{3.11}$$

$$\implies \Lambda_\mu(q) = \gamma^0\Lambda^\dagger_\mu(-q)\gamma^0. \tag{3.12}$$

Substituting (3.12) into (3.9) constraints the form factor functions such that

$$\mathbb{f}_2,\ \mathbb{f}_3,\ \mathbb{f}_4 \text{ are real} \tag{3.13}$$

and

$$\mathbb{f}_1,\ \mathbb{f}_5,\ \mathbb{f}_6 \text{ are imaginary.} \tag{3.14}$$

Furthermore, imposing current conservation $\partial^\mu j^{(\nu)}_\mu = 0$ required by the EM $U(1)$ gauge invariance on $A^\mu \to A^\mu + \partial^\mu\varphi(x)$ for any $\varphi(x)$ which leaves (2.2) invariant in combination with (3.4) leads to

$$\langle \nu(p_f)|\left[\mathcal{P}^\mu, j^{(\nu)}_\mu(0)\right]|\nu(p_i)\rangle = 0 \tag{3.15}$$

$$\implies q^\mu\bar{u}(p_f)\Lambda_\mu(q)u(p_i) = 0. \tag{3.16}$$

Utilizing on-shell conditions for the neutrino spinors $\not{p}_{i(f)}u(p_{i(f)}) = m\,u(p_{i(f)})$, the antisymmetry of $\sigma_{\mu\nu}$ and $\epsilon_{\mu\nu\rho\gamma}$ and the anticommutation relation of γ_μ with γ_5

$$\begin{aligned}
q^\mu \bar{u}(p_f)\Lambda_\mu(q)u(p_i) &= \bar{u}(p_f)[\mathbb{f}_1(q^2)\, q^\mu q_\mu + \mathbb{f}_2(q^2)\, q^\mu q_\mu \gamma_5 + \mathbb{f}_3(q^2)\, \overbrace{q^\mu \gamma_\mu}^{\not{q}=\not{p}_i-\not{p}_f} \\
&\quad + \mathbb{f}_4(q^2)\, \underbrace{q^\mu \gamma_\mu}_{\not{q}=\not{p}_i-\not{p}_f}\, \gamma_5 + \mathbb{f}_5(q^2)\, \cancelto{0}{q^\mu \sigma_{\mu\nu} q^\nu} + \mathbb{f}_6(q^2)\, \cancelto{0}{q^\mu \epsilon_{\mu\nu\rho\lambda} q^\nu} \sigma^{\rho\lambda}]u(p_i) \\
&= \bar{u}(p_f)[\mathbb{f}_1(q^2)\, q^2 + \mathbb{f}_2(q^2)\, q^2 \gamma_5 \\
&\quad + \mathbb{f}_3(q^2)\, (\not{p}_i - \not{p}_f) + \mathbb{f}_4(q^2)\, (\not{p}_i - \not{p}_f)\gamma_5]u(p_i) \\
&= \bar{u}(p_f)[\mathbb{f}_1(q^2)\, q^2 + \mathbb{f}_2(q^2)\, q^2 \gamma_5 \\
&\quad + \mathbb{f}_3(q^2)\, \cancelto{0}{(m - m)} + \mathbb{f}_4(q^2)\, (m + m)\gamma_5]u(p_i) \\
&= \bar{u}(p_f)[\mathbb{f}_1(q^2)\, q^2 + \mathbb{f}_2(q^2)\, q^2 \gamma_5 + 2m\, \mathbb{f}_4(q^2)\gamma_5]u(p_i). \qquad (3.17)
\end{aligned}$$

Substituting (3.17) into (3.16)

$$\implies \mathbb{f}_1(q^2)\, q^2 + \mathbb{f}_2(q^2)\, q^2 \gamma_5 + 2m\, \mathbb{f}_4(q^2)\gamma_5 = 0. \qquad (3.18)$$

Taking into account that $\mathbb{I}$ and γ_5 are linearly independent leaves $\mathbb{f}_1(q^2) = 0$ and $\mathbb{f}_4(q^2) = -\frac{q^2}{2m}\mathbb{f}_2(q^2)$. Substituting back into (3.9) leaves

$$\Lambda_\mu(q) = \mathbb{f}_Q(q^2)\gamma_\mu - \mathbb{f}_M(q^2) i\sigma_{\mu\nu}q^\nu + \mathbb{f}_E(q^2)\sigma_{\mu\nu}q^\nu\gamma_5 + \mathbb{f}_A(q^2)(q^2\gamma_\mu - q_\mu \not{q})\gamma_5, \qquad (3.19)$$

where $\mathbb{f}_Q = \mathbb{f}_3$, $\mathbb{f}_M = i\mathbb{f}_5$, $\mathbb{f}_E = -2i\mathbb{f}_6$ and $\mathbb{f}_A = -\mathbb{f}_2/2m$ are the real charge, dipole magnetic and electric, and anapole neutrino form factors. For the coupling with the real photon at $q^2 = 0$

$$\mathbb{f}_Q(0) = \mathbb{q}, \qquad \mathbb{f}_M(0) = \mu, \qquad \mathbb{f}_E(0) = \epsilon, \qquad \mathbb{f}_A(0) = \mathbb{a}, \qquad (3.20)$$

where $\mathbb{q}$ is the electric charge, μ the magnetic dipole, ϵ the electric dipole and $\mathbb{a}$ the anapole moment of the neutrino.

3.2.2 For Mixed Massive Neutrinos

It is worth noting that we have only considered the effective interaction of a single mass eigenstate neutrino with the photon, whereas a full treatment of the known mass mixing among neutrinos requires promoting the aforementioned vertex function and form factors into $N \times N$ matrices, where $\{|\nu_k\rangle\}$, $k = 1, 2, \cdots, N$, describe N neutrino mass eigenstates of definite mass satisfying $(i\not{\partial} - m_k)\nu_k = 0$. A generalized version of (3.2), described by the effective Feynman diagram in Fig. 3.1(c), is given by

$$\mathcal{H}^{(\nu)}_{\text{EM}}(x) = j^{(\nu)}_\mu(x)A^\mu(x) = \sum_{k,j=1}^{N} \overline{\nu_k}(x)\Lambda^{kj}_\mu \nu_j(x)A^\mu(x), \qquad (3.21)$$

where, for an initial state $|\nu_i(p_i)\rangle$ and a final state $|\nu_f(p_f)\rangle$ describing free mass-eigenstate neutrinos, the generalized form of the matrix element in (3.8) reads

$$\langle \nu_f(p_f)|\, j_\mu^{(\nu)}(0)\, |\nu_i(p_i)\rangle = \overline{\nu_f}(p_f)\, \Lambda_\mu^{fi}(p_f, p_i)\, \nu_i(p_i). \tag{3.22}$$

As was the case for $\Lambda_\mu(p_f, p_i)$, the matrix $\Lambda_\mu^{fi}(p_f, p_i)$ depends only on q for the same underlying reason, and in its most general form is a linear combination of the same covariant elements, namely

$$\begin{aligned}\Lambda_\mu^{fi} &= \mathbb{f}_1^{fi}(q^2)\, q_\mu + \mathbb{f}_2^{fi}(q^2)\, q_\mu\gamma_5 + \mathbb{f}_3^{fi}(q^2)\, \gamma_\mu \\ &\quad + \mathbb{f}_4^{fi}(q^2)\, \gamma_\mu\gamma_5 + \mathbb{f}_5^{fi}(q^2)\, \sigma_{\mu\nu}q^\nu + \mathbb{f}_6^{fi}(q^2)\, \epsilon_{\mu\nu\rho\lambda}\, q^\nu\sigma^{\rho\lambda},\end{aligned} \tag{3.23}$$

with conditions (3.13) and (3.14) holding for the corresponding $\mathbb{f}_m^{fi}$, $m = 1, \cdots, 6$. Imposing current conservation, the on-shell conditions for the neutrino spinors $\overline{\nu_f}\not{q}\nu_i = \overline{\nu_f}(m_f - m_i)\nu_i$ and the algebraic properties of the covariant forms appearing in (3.23) yields the constraints

$$\begin{cases} \mathbb{f}_1^{fi}(q^2) + (m_f - m_i)\mathbb{f}_3^{fi}(q^2) = 0, & (3.24)\\ \mathbb{f}_2^{fi}(q^2) + (m_f + m_i)\mathbb{f}_4^{fi}(q^2) = 0. & (3.25)\end{cases}$$

so that the vertex function (3.19) is correspondingly promoted to

$$\Lambda_\mu^{fi}(q) = (\gamma_\mu - q_\mu\not{q}/q^2)(\mathbb{f}_Q^{fi}(q^2) + \mathbb{f}_A^{fi}(q^2)q^2\gamma_5) - i\sigma_{\mu\nu}q^\nu(\mathbb{f}_M^{fi}(q^2) + i\mathbb{f}_E^{fi}(q^2)\gamma_5), \tag{3.26}$$

where $\mathbb{f}_Q^{fi} = \mathbb{f}_3^{fi}$, $\mathbb{f}_M^{fi} = i\mathbb{f}_5^{fi}$, $\mathbb{f}_E^{fi} = -2i\mathbb{f}_6^{fi}$, and $\mathbb{f}_A^{fi} = -\mathbb{f}_2^{fi}/(m_f + m_i)$, subject to the Hermiticity condition

$$\mathbb{f}_\Omega^{fi} = (\mathbb{f}_\Omega^{if})^* \qquad (\Omega = Q, E, M, A). \tag{3.27}$$

Coupling with the real photon at $q^2 = 0$

$$\mathbb{f}_Q^{fi}(0) = \mathbb{q}^{fi}, \qquad \mathbb{f}_M^{fi}(0) = \mu^{fi}, \qquad \mathbb{f}_E^{fi}(0) = \epsilon^{fi}, \qquad \mathbb{f}_A^{fi}(0) = \mathbb{a}^{fi}, \tag{3.28}$$

where $\mathbb{q}^{fi}$ is the electric charge, μ^{fi} the magnetic dipole, ϵ^{fi} the electric dipole and $\mathbb{a}^{fi}$ the anapole transition moment of an initial $|\nu_i\rangle$ and final $|\nu_f\rangle$ neutrino mass eigenstate.

It is worth noting that the diagonal entries $f = i$ of $\mathbb{f}_\Omega^{fi}$ and $\Lambda_\mu^{fi}(q)$ reduce to their single mass-eigenstate counterparts found previously, whereas the off-diagonal entries $f \neq i$ describe transition electromagnetic properties between distinct mass eigenstates, a feature entirely absent in the single-flavor treatment and one of the principal new phenomena introduced by neutrino mixing.

3.2.3 For Neutrinos in Flavor Base

It is important to recognize, however, that the mass eigenstates $|\nu_i\rangle$ are not the states that are produced or detected in weak interaction processes; rather, it is the flavor eigenstates $|\nu_\alpha\rangle$, $\alpha = e, \mu, \tau$, that couple directly to the charged leptons through the $W^\pm$ boson, the two bases being related by the leptonic mixing matrix U through $|\nu_\alpha\rangle = \sum_i U^*_{\alpha i} |\nu_i\rangle$, better known as the PMNS matrix [25, 26]. Consequently, the electromagnetic vertex function in the flavor basis, $\widetilde{\Lambda}^{\alpha\beta}_\mu$, is obtained from Λ^{ij}_μ by the corresponding unitary rotation

$$\widetilde{\Lambda}^{\alpha\beta}_\mu(q) = \sum_{i,j=1}^{3} U_{\alpha i}\, \Lambda^{ij}_\mu(q)\, U^*_{\beta j}, \tag{3.29}$$

with the inverse relation $\Lambda^{ij}_\mu = \sum_{\alpha,\beta} U^*_{\alpha i}\, \widetilde{\Lambda}^{\alpha\beta}_\mu\, U_{\beta j}$ following immediately from the unitarity of U. While the algebraic structure of (3.26) is preserved under this rotation and thus the EM form factors can be definned in the flavor basis such that

$$\widetilde{f}^{\alpha\beta}_\Omega(q^2) = \sum_{i,j=1}^{3} U_{\alpha i}\, \mathbb{f}^{ij}_\Omega(q^2)\, U^*_{\beta j}, \tag{3.30}$$

the flavor basis offers no particular simplification on its own; rather, its utility becomes apparent once the calculation of the SM electroweak loop diagrams generating Λ_μ is undertaken, since it is precisely in this basis that the relevant interaction vertices are diagonal.

3.2.4 For Weyl Neutrinos

A further simplification arises upon considering the strictly massless limit[2], in which the neutrino mass eigenstates reduce to two-component Weyl fermions and helicity coincides exactly with chirality. In this limit, the physical neutrino exists only as the left-handed projection $\nu_{\alpha,\beta(i,j)L}$, while the corresponding antineutrino is purely right-handed, and no Lorentz transformation can connect the two. Since the bilinears $\sigma_{\mu\nu}q^\nu$ and $\sigma_{\mu\nu}q^\nu\gamma_5$ appearing in (3.26) are odd under chirality and therefore connect left- to right-handed spinors, their matrix elements between same-chirality Weyl states vanish identically, namely $\overline{\nu_{\alpha(i)L}}\, \sigma_{\mu\nu}q^\nu\, \nu_{\beta(j)L} = \overline{\nu_{\alpha(i)L}}\, \sigma_{\mu\nu}q^\nu\gamma_5\, \nu_{\beta(j)L} = 0$, so that $\widetilde{f}^{\alpha\beta}_M$ ($\mathbb{f}^{ij}_M$) and $\widetilde{f}^{\alpha\beta}_E$ ($\mathbb{f}^{ij}_E$) are forced to vanish in the massless limit. The vertex function correspondingly collapses to

$$\widetilde{\Lambda}^{\alpha\beta}_\mu(q)\Big|_{m_\nu=0} = (\gamma_\mu - q_\mu \not{q}/q^2)\left(\widetilde{f}^{\alpha\beta}_Q(q^2) - \widetilde{f}^{\alpha\beta}_A(q^2)q^2\gamma_5\right), \tag{3.31}$$

leaving the charge and anapole form factors as the only surviving electromagnetic structures available to a Weyl neutrino, consistent with the well-known statement that the magnetic and electric dipole moments of a Dirac neutrino are proportional to its mass and hence vanish identically in this limit. Moreover, since the one-loop SM diagrams contributing to the vertex function in the massless limit involve only same-flavor charged lepton propagators running in the loop, the vertex function is

[2] We will show in later sections that this is consistent with the SM and a good approximation at the energy thresholds considered in this work.

diagonal in the flavor basis: $\widetilde{\Lambda}_{\mu}^{\alpha\beta} = 0$ and $\widetilde{f}_{Q,A}^{\alpha\beta} = 0$ for $\alpha \neq \beta$ and each of the diagonal entries correspond to electromagnetic properties specific to each neutrino flavor.

3.3. Neutrino Charge Radius

Even if the neutrino is of neutral net charge, its charge form factor $f_Q(q^2)$ can still hold information as to its interaction with other charged particles. In the "Briet frame", defined by the time coordinate of four-momentum transfer such that $q_0 = 0$, $f_Q(q^2)$ only depends on $|\vec{q}| = \sqrt{-q^2}$ and can be interpreted as the Fourier transform of a spherically symmetric charge distribution $\rho(r)$ where $r = |\vec{r}|$.

$$f_Q(q^2) = \int \rho(r) e^{-i\vec{q}\cdot\vec{r}} d^3\vec{r} = \int \rho(r) \frac{\sin(|\vec{q}|r)}{|\vec{q}|r} d^3\vec{r}. \tag{3.32}$$

Deriving with respect to $q^2 = -|\vec{q}|^2$,

$$\frac{df_Q}{dq^2} = \int \rho(r) \frac{\sin(|\vec{q}|r) - |\vec{q}|\cos(|\vec{q}|r)}{2q^3 r} d^3\vec{r}, \tag{3.33}$$

and thus taking

$$\lim_{q^2 \to 0} \frac{df_Q}{dq^2} = \int \rho(r) \frac{r^2}{6} d^3\vec{r} = \frac{\langle r^2 \rangle}{6}. \tag{3.34}$$

Therefore, the average of the squared radius of the neutrino charge distribution is given by

$$\langle r^2 \rangle = 6 \left. \frac{df_Q}{dq^2} \right|_{q^2=0}. \tag{3.35}$$

Expanding $f_Q(q^2)$ around $q^2 = 0$ in a Taylor series yields

$$\begin{aligned} f_Q(q^2)\big|_{q\approx 0} &\approx \cancelto{0}{f_Q(0)} + \left. \frac{df_Q}{dq^2} \right|_{q^2=0} q^2 \\ &\approx \frac{\langle r^2 \rangle}{6} q^2, \end{aligned} \tag{3.36}$$

and substituting (3.36) into a matrix element of a conserved current with a vertex function of the from in (3.31) at the same limit gives

$$\begin{aligned} \overline{\nu_L}(p_f)\Lambda_{\mu}^{\mathrm{SM}}(q)\nu_L(p_i)\Big|_{q^2\approx 0} &= \overline{\nu_L}(p_f)(\gamma_\mu - \cancelto{0}{q_\mu \not{q}/q^2})(f_Q(q^2) + f_A(q^2)q^2\gamma_5)\nu_L(p_i), \\ &\approx \overline{\nu_L}(p_f)(\gamma_\mu)(\frac{\langle r^2 \rangle}{6} - \mathbb{a})q^2 \nu_L(p_i), \end{aligned} \tag{3.37}$$

utilizing $\gamma_5 \nu_L = -\nu_L$, $\overline{\nu_L}(p_f)\not{q}\nu_L(p_i) = 0$, and $f_A(0) = \mathbb{a}$. Because of the from of $\Lambda_{\mu}^{\mathrm{SM}}$ implied in (3.37) it's actually a matter of convention between authors what gets defined as the NCR since defining the total neutrino from factor $F_\nu(q^2)$ such that

$$\Lambda_{\mu}^{\mathrm{SM}}(q) = \gamma_\mu F_\nu(q^2), \tag{3.38}$$

where $F_\nu(q^2) = f_Q(q^2) + q^2 f_A(q^2)$ acts as such that a SM $\nu\nu\gamma$ effective coupling cannot distinguish between an anapole and a charge from factor contribution at the $q \to 0$ limit and thus $F_\nu(q^2)\Big|_{q^2\to 0} = f_Q(q^2)\Big|_{q^2\to 0} \implies \text{a} = -\langle r^2\rangle/6$ for ultra-relativistic left-hand neutrinos. This results in a redefining of an average square charge radius not in terms of f_Q, but rather F_ν as a whole which we will denote $\langle r^2\rangle^{\text{SM}}$

$$\Lambda_\mu^{\text{SM}}(q) = \gamma_\mu F_\nu(q^2) \simeq \gamma_\mu\, q^2 \frac{\langle r^2\rangle^{\text{SM}}}{6} \simeq \gamma_\mu\, q^2 \frac{\langle r^2\rangle}{3}. \tag{3.39}$$

Different authors report either $\langle r^2\rangle$ or $\langle r^2\rangle^{\text{SM}}$ and thus the literature contains differences of factors of 2 for the NCR as seen from (3.39). The usual convention for contemporary neutrino phenomenology, and the one that will be used in this work, is to define

$$\langle r^2\rangle^{\text{SM}} = +6 \frac{dF_\nu(q^2)}{dq^2}\Big|_{q=0}, \tag{3.40}$$

and report $\langle r^2\rangle^{\text{SM}}$ as the NCR.

3.4. SM Corrections to the $\nu\nu\gamma$ vertex

The SM prediction for $\langle r^2\rangle_{\alpha\alpha} = \langle r^2_{\nu_\alpha}\rangle$ as defined by

$$\langle r^2_{\nu_\alpha}\rangle = +6 \frac{d\widetilde{f}^{\alpha\alpha}}{dq^2}\Big|_{q^2=0} \tag{3.41}$$

arises at one loop from radiative corrections to the $\nu\nu\gamma$ vertex. Within the pinch technique (PT) framework of [27], one constructs a gauge-invariant, process-independent effective vertex $\hat{\Gamma}^\mu_{\gamma\nu\bar{\nu}}$ by systematically extracting all longitudinal momenta from the conventional vertex, box, and self-energy diagrams and reassigning the resulting propagator-like pieces via the elementary Ward identities of the theory. The diagrams that contribute to $\hat{\Gamma}^\mu_{\gamma\nu\bar{\nu}}$ after this rearrangement are those in which a virtual W boson carries the photon coupling while the associated charged lepton ℓ_α runs in the loop; the remaining diagrams either cancel in the PT procedure or are absorbed into the effective self-energies $\hat{\Pi}^{\mu\nu}_{gZ}$ and $\hat{\Pi}^{\mu\nu}_{ZZ}$, which are instead interpreted as process-independent corrections to the effective weak mixing angle [27]. Crucially, the W vertex is diagonal in flavor, it couples ν_α exclusively to ℓ_α, so the loop integral depends only on m_{ℓ_α} and M_W, and the resulting charge form factor $\widetilde{f}^\alpha_Q(q^2)$ is diagonal in the flavor basis. The Dirac form factor extracted from $\hat{\Gamma}^\mu_{\gamma\nu\bar{\nu}}$ then yields, in the limit $q^2 \to 0$[3] [27],

$$\langle r^2_{\nu_\alpha}\rangle = \frac{\langle r^2_{\nu_\alpha}\rangle^{\text{SM}}}{2} = -\frac{G_F}{4\sqrt{2}\,\pi^2}\left[3 - 2\ln\frac{m^2_{\ell_\alpha}}{M^2_W}\right]. \tag{3.42}$$

which is finite, gauge-independent, and depends only on the flavor index α through the charged lepton mass m_α.

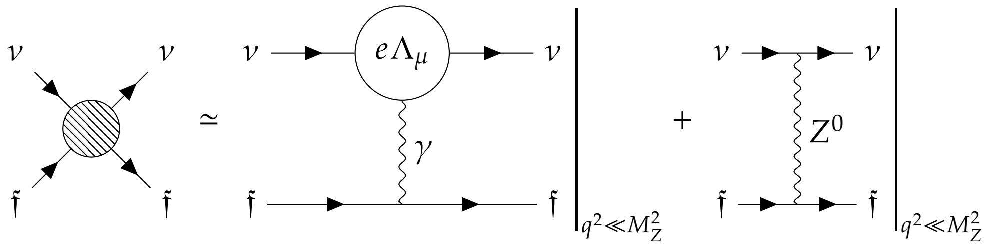
[3] utilizing the definition in (3.40)

Numerically, the values of NCR prediction for each known neutrino flavour in the SM are

$$\langle r^2_{\nu_e}\rangle^{\text{SM}} = -0.83\times 10^{-32}\,\text{cm}^2, \tag{3.43}$$
$$\langle r^2_{\nu_\mu}\rangle^{\text{SM}} = -0.48\times 10^{-32}\,\text{cm}^2, \tag{3.44}$$
$$\langle r^2_{\nu_\tau}\rangle^{\text{SM}} = -0.30\times 10^{-32}\,\text{cm}^2. \tag{3.45}$$

3.5. NCR as a radiative correction of Electroweak Mixing

With a SM prediction for the effective $\nu\nu\gamma$ vertex now in hand, it is possible to include its contribution in the calculation of a neutral current interaction amplitude between a SM left-handed neutrino and a generic fermion $\mathfrak{f}$ of non-zero electric charge, in the limit $q^2 \ll M_Z^2$. Utilizing the vertices in (2.3), (2.42), where for a left-handed neutrino $g_V^\nu = g_A^\nu = 1/2$, and (3.39), now written with the explicit dependence on the electromagnetic coupling constant e that was omitted in Sec. 3.2, as well as the propagators (2.14) with $\xi = 1$ and (2.20),



(3.46)

$$\begin{aligned}\Longrightarrow\; -i\mathcal{M} \simeq &\left[-ie\overline{\nu_L}\gamma_\mu\, q^2\,\frac{\langle r_\nu^2\rangle^{\text{SM}}}{6}\nu_L\right]\frac{-i\eta^{\mu\nu}}{q^2}\left[\bar{\mathfrak{f}} - ieQ_{\mathfrak{f}}\gamma_\nu\mathfrak{f}\right]\\ &+\left[\frac{-ig}{2\cos\theta_W}\overline{\nu_L}\gamma_\mu\Big(\frac{1}{2}-\frac{1}{2}\gamma_5\Big)\nu_L\right]\frac{+i\eta^{\mu\nu}}{M_Z^2}\left[\frac{-ig}{2\cos\theta_W}\bar{\mathfrak{f}}\gamma_\mu(g_V^{\mathfrak{f}} - g_V^{\mathfrak{f}}\gamma_5)\mathfrak{f}\right].\end{aligned} \tag{3.47}$$

One can observe from (3.47) that the q^2 dependence of Λ_μ cancels exactly against the photon propagator, so that even in the low-energy limit the electromagnetic form factors survive. Taking into account that $\nu_L = \frac{1}{2}(1-\gamma_5)\nu_L$, together with (2.40) and (2.50), further simplifications yield

$$\begin{aligned}\mathcal{M} = &\underbrace{\left[e\overline{\nu_L}\gamma_\mu\nu_L\right]}_{=\overline{\nu_L}e\gamma_\mu\frac{1}{2}(1-\gamma_5)\nu_L}\left[\bar{\mathfrak{f}}\frac{e\langle r_\nu^2\rangle^{\text{SM}}Q_{\mathfrak{f}}}{6}\gamma^\mu\mathfrak{f}\right] - \underbrace{\frac{g^2}{8\cos^2\theta_W M_Z^2}}_{=\frac{g^2}{8M_W^2}=\frac{G_F}{\sqrt{2}}}\left[\overline{\nu_L}\gamma_\mu(1-\gamma_5)\nu_L\right]\left[\bar{\mathfrak{f}}(g_V^{\mathfrak{f}} - g_A^{\mathfrak{f}}\gamma_5)\mathfrak{f}\right],\\ = &\frac{G_F}{\sqrt{2}}\left[\overline{\nu_L}\gamma_\mu(1-\gamma_5)\nu_L\right]\left[\bar{\mathfrak{f}}\gamma^\mu\left(\left(g_V^{\mathfrak{f}} - \frac{\sqrt{2}e^2Q_{\mathfrak{f}}\langle r_\nu^2\rangle^{\text{SM}}}{12\,G_F}\right) - g_A^{\mathfrak{f}}\gamma_5\right)\mathfrak{f}\right].\end{aligned} \tag{3.48}$$

Eq. (3.48) demonstrates explicitly that the inclusion of the effective $\nu\nu\gamma$ vertex is equivalent to a Fermi-like neutral current interaction with a shifted vector coupling of the target fermion. Writing $e^2 = 4\pi\alpha_{\text{EM}}$, the shift takes the form

$$g_V^{\mathfrak{f}} \to g_V^{\mathfrak{f}} - \frac{\sqrt{2}\, Q_{\mathfrak{f}} \pi \alpha_{\text{EM}}}{3\, G_F} \langle r_\nu^2 \rangle^{\text{SM}}. \tag{3.49}$$

Eq. (3.49) can equivalently be expressed as a shift in $\sin^2\theta_W$ by making use of (2.32) and (2.40),

$$\begin{aligned}
g_V^{\mathfrak{f}} \to g_V^{\mathfrak{f}} & - \frac{\sqrt{2} e^2 Q_{\mathfrak{f}} \langle r_\nu^2 \rangle^{\text{SM}}}{12\, G_F} \\
&= T_{\mathfrak{f}}^3 - 2Q_{\mathfrak{f}} \sin^2\theta_W + \frac{1}{12}\frac{\sqrt{2}}{G_F} e^2 Q_{\mathfrak{f}} \langle r_\nu^2 \rangle^{\text{SM}} \\
&= T_{\mathfrak{f}}^3 - 2Q_{\mathfrak{f}} \left(\sin^2\theta_W + \frac{1}{24}\frac{8M_W^2}{g^2} (g^2 \sin^2\theta_W) \langle r_\nu^2 \rangle^{\text{SM}} \right) \\
&= T_{\mathfrak{f}}^3 - 2Q_{\mathfrak{f}} \left(\sin^2\theta_W + \frac{1}{3} M_W^2 \sin^2\theta_W \langle r_\nu^2 \rangle^{\text{SM}} \right) \\
&= T_{\mathfrak{f}}^3 - 2Q_{\mathfrak{f}} \sin^2\theta_W \left(1 + \frac{1}{3} M_W^2 \langle r_\nu^2 \rangle^{\text{SM}} \right)
\end{aligned} \tag{3.50}$$

Eq. (3.50) reveals that the (3.49) shift is equivalent to

$$\sin^2\theta_W \to \sin^2\theta_W \left(1 + \frac{1}{3} M_W^2 \langle r_\nu^2 \rangle^{\text{SM}} \right) \tag{3.51}$$

Eq. (3.51) will be how the antineutrino charge radius will be incorporated to the interaction cross-section used in the following chapters of this work.

4. Coherent Elastic Neutrino-Nucleus Scattering

This chapter introduces the main interaction by which the sensitivities of neutrino detectors to the SM parameters are determined, describing a neutral current interaction along with its cross section, and showing how, in conjunction with a model of the antineutrino flux from a reactor source, the expected number of events for a broadly-described experimental design can be calculated; this number will be used in later chapters to test how different values of the SM parameters influence it.

4.1. Introduction

Coherent Elastic Neutrino-Nucleus Scattering (CEνNS) is a neutral current weak interaction within the SM originally proposed in 1979 [28] and measured successfully in 2017 [29], wherein low-energy neutrinos [$E_\nu \sim$ MeV] collide with atomic nuclei as coherent bodies rather than their constituents, imparting on them a small recoil energy. These interactions are possible with neutrino energies up to $50\,MeV$ with medium-sized nuclei, since their main condition consists of the momentum transfer, Q, being negligible in comparison with the nuclear radius, R ($QR \ll 1$) [30]. The effective Feynman diagram for these interactions is shown in Fig. 4.1.

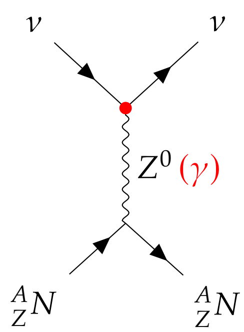


Figure 4.1: *Tree-level Feynmann diagram for CEνNS interaction between a neutrino ν and a nucleus N with nucleon number A and proton number Z*

The calculation of its differential cross-section in the low-energy limit consists of setting an effective Lagrangian density given by:

$$\mathcal{L}_{\text{eff}} = \frac{G}{\sqrt{2}} j^{(\nu)\mu} \mathcal{J}_\mu, \tag{4.1}$$

where $j^{(\nu)\mu}$ denotes a neutral neutrino current and $\mathcal{J}_\mu$ a hadronic current from which the nuclear mass distribution terms originate.

4.2. Cross Section

The low-energy differential cross-section in terms of the nuclear recoil, E_R, is given by [31]:

$$\frac{d\sigma}{dT}(E_\nu, E_R) = \frac{G_F^2 M}{\pi} Q_W^2 \left(1 - \frac{M E_R}{2E_\nu}\right), \tag{4.2}$$

where M is the target nuclear mass and Q_W^2 is the vector nuclear weak charge given in terms of the proton (p) and neutron (n) weak couplings:

$$Q_W^2 = [Z g_V^p F_p(q^2) + N g_V^n F_n(q^2)]^2, \tag{4.3}$$

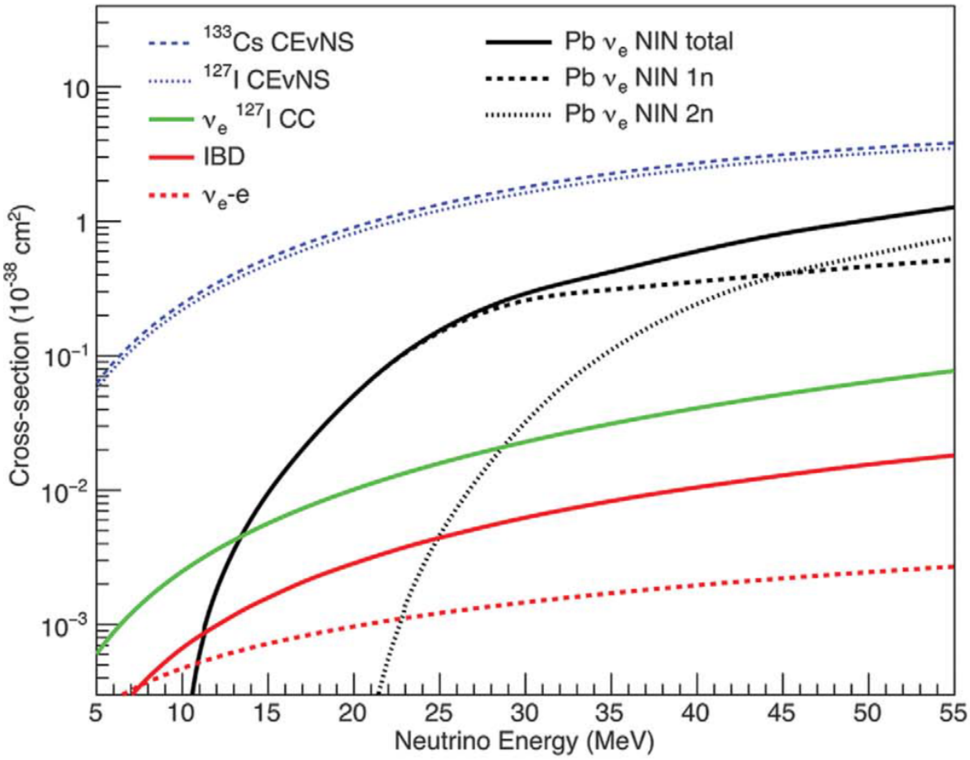


Figure 4.2: *Total cross sections from CEνNS and some known neutrino couplings. Included are neutrino-electron scattering, charged-current (CC) interaction with iodine, inverse beta decay (IBD) and neutrino-induced neutron (NIN) generation for 1n and 2n [29]*

where g_V^p and g_V^n are given by (2.44) in the specific cases for the proton and neutron respectively:

$$g_V^p = T_p^3 - 2\sin^2(\theta_W)Q_p = \frac{1}{2} - 2\sin^2(\theta_W), \tag{4.4}$$

$$g_V^n = T_n^3 - 2\sin^2(\theta_W)Q_n = -\frac{1}{2}, \tag{4.5}$$

and $F_{n,p}(Q^2)$ are the neutron and proton nuclear form factors, dependent on the momentum transfer $q^2 = 2MT$. In Eqs. (4.3), (4.4), and (4.5), Z is the atomic number, $N = A - Z$ is the neutron number, and A is the mass (nucleon) number of the target nucleus. Fig. 4.2 present a graph for the CEνNS cross section compared to other neutrino interactions. It is worth noting that, given the weak mixing angle $\sin^2(\theta_W) \approx 0.23156 \approx \frac{1}{4}$[32], Q_W^2 (and by extension $\frac{d\sigma}{dT}$) is mostly $\propto N^2$. As discussed in Chapter 3, the inclusion of the neutrino charge radius (NCR) correction comes in the form of considering the shift in the $g_V^{\mathfrak{f}}$ couplings for the fermions $\mathfrak{f} = p, n$. Crucially, since $Q_n = 0$, the shift is only present in the proton vector coupling. Substituting $\mathfrak{f} = p$ and $Q_p = +1$ into (3.49) gives:

$$g_V^p \rightarrow g_V^p - \frac{\sqrt{2}\pi\alpha_{\text{EM}}}{3\,G_F}\langle r_\nu^2\rangle^{\text{SM}}, \tag{4.6}$$

which as already discussed is equivalent to $\sin^2\theta_W \rightarrow \sin^2\theta_W\left(1 + \frac{1}{3}M_W^2\langle r_\nu^2\rangle^{\text{SM}}\right)$.

	^{235}U	^{238}U	^{239}Pu	^{241}Pu
α_1	3.217	$4.833\cdot10^{-1}$	6.413	3.251
α_2	-3.111	$1.927\cdot10^{-1}$	-7.432	-3.204
α_3	1.395	$-1.283\cdot10^{-1}$	3.535	1.428
α_4	$-3.690\cdot10^{-1}$	$-6.762\cdot10^{-3}$	$-8.820\cdot10^{-1}$	$-3.675\cdot10^{-1}$
α_5	$4.445\cdot10^{-2}$	$2.233\cdot10^{-3}$	$1.025\cdot10^{-1}$	$4.254\cdot10^{-2}$
α_6	$-2.053\cdot10^{-3}$	$-1.536\cdot10^{-4}$	$-4.550\cdot10^{-3}$	$-1.869\cdot10^{-3}$

Table 4.1: *Experimentally fitted coefficients α_{kp} as published in [40].*

4.3. Neutrino Sources

Part of the calculation of the expected number of CEνNS events is the energy-dependent flux of incoming (anti)neutrinos. This work focuses mainly on a model for electron-antineutrinos known as the Huber-Mueller model.

4.3.1 Nuclear Reactors

Ever since their discovery in 1959 [2], reactor neutrino sources have been fundamental for neutrino physics. In recent times they have still been utilized in experiments such as the measurement of Δm_{21} in KamLAND [33] and of the mixing angle θ_{13} in Daya Bay [34]. The main inspiration for the present work comes from two papers, one by L. J. Flores et al. [35] and another by E. Alfonso-Pita et al. [36], wherein a 10 kg scintillating liquid Argon (LAr) bubble chamber detector is placed at a distance of ~3 m from a reactor running at ~1 MW$_{th}$ thermal power, based on a TRIGA Mark III nuclear reactor at the National Institute of Nuclear Research (ININ) facilities on the Mexico-Toluca highway near Mexico City.

Nuclear reactors commonly operate based on Uranium and Plutonium, whose fission produces a large quantity of neutron-rich remnant nuclei that subsequently undergo beta decay. The resulting antineutrino flux for ^{235}U, ^{239}Pu, and ^{241}Pu has been calculated by the Institut Laue-Langevin (ILL) in the 1980s [37, 38]. Coupled with an *ab initio* method for the estimation of the ^{238}U spectrum as seen in [39], the following model is obtained for the total electron antineutrino spectrum, known as the Huber-Mueller model [40]:

$$\frac{d\Phi}{dE_\nu}(E_\nu) = \frac{P}{4\pi d^2} \sum_k \frac{f_k}{\epsilon_k} \exp\left(\sum_{p=1}^{6} \alpha_{kp} E_\nu^{p-1} \right), \tag{4.7}$$

where P is the thermal power of the reactor, d is the reactor-detector distance, $k \in \{^{235}\text{U},^{239}\text{Pu},^{241}\text{Pu},^{238}\text{U}\}$ is an index that runs through the contribution of the isotopes involved, ϵ_k is the energy per fission of each isotope as published in [41], f_k is the fission fraction of each isotope, and α_{kp} are the experimentally fitted coefficients shown in Table 4.1. In Fig. 4.3 the individual isotope contributions are plotted against each other in arbitrary units, and Fig. 4.4 shows the total flux given the fission fractions: 55% of ^{235}U, 32% of ^{239}Pu, 6% of ^{241}Pu, and 7% of ^{238}U, in accordance with [42].

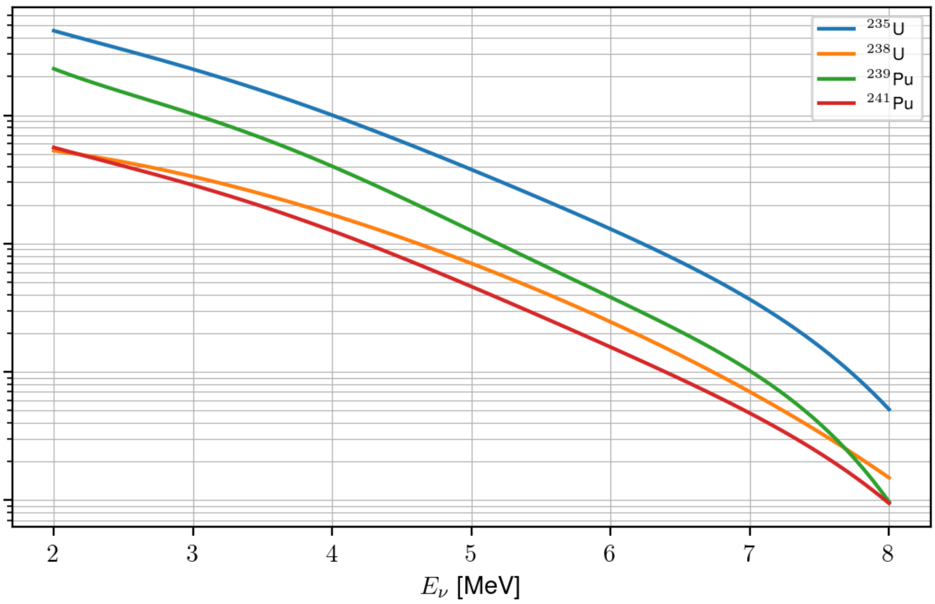


Figure 4.3: *Isotope contributions to total antineutrino flux spectrum.*

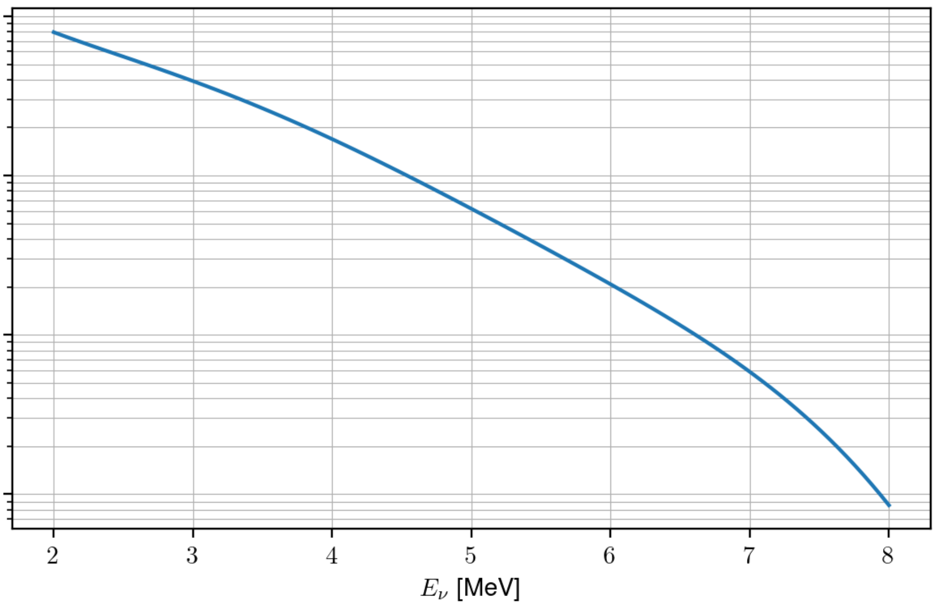


Figure 4.4: *Total antineutrino flux spectrum.*

4.4. Expected Number of Events

Ignoring reactor resolution and efficiency, the expected number of events of a detector of mass M over an exposure time t is given by

$$N = t\,\frac{M_{\text{Detector}}}{M}\int_{E_{R\,\text{th}}}^{E_{R\,\text{max}}} dE_R \int_{E_{\nu,\text{min}}(E_R)}^{E_{\nu,\text{max}}} dE_\nu \frac{d\Phi}{dE_\nu}(E_\nu)\left(\frac{d\sigma}{dE_R}\right), \tag{4.8}$$

where $E_{R\,\text{th}}$ a threshold nuclear recoil energy, $E_{R\,\text{max}}$ a kinematically determined maximum recoil energy, $E_{\nu,\text{max}} \sim 9.5$ MeV the maximum incident neutrino energy given by the neutrino spectrum [38] and $E_{\nu,\text{min}}(E_R) \approx \sqrt{ME_R/2}$ the minimum incident neutrino energy required to produce a given E_R recoil energy

$$E_{\nu,\text{min}}(E_R) = \frac{E_R}{2}\left(1 + \sqrt{1 + \frac{2M}{E_R}}\right). \tag{4.9}$$

While calculations derived from these definitions can already be sufficient for probing sensitivities in neutrino detection experiments, considerations pertaining to actual scintillation experiments can be accounted for in the form of a quenching factor (QF), which suppresses the actual measured amount of CEνNS events in a realistic experiment.

4.4.1 Quenching Factor

The differential cross section $\frac{d\sigma}{dE_R}$ and the event rate integral in (4.8) are both expressed in terms of the true nuclear recoil energy, E_R, transferred to the target nucleus. However, detectors such as scintillating LAr chambers do not measure E_R directly; instead, they measure an *ionization* (or scintillation) energy, E_I, produced by the recoiling nucleus as it deposits its energy in the medium. The two quantities are related by the quenching factor (QF), $Q(E_R)$, defined through

$$E_I = Q(E_R)E_R. \tag{4.10}$$

This factor is incorporated into binned analyses of event counts through the substitution [43]

$$\int_{E_R(i)}^{E_R(i+i)} dE_R = \int_{E_I(i)}^{E_I(i+i)} \frac{1}{Q(E_I)}\left(1 - \frac{E_I}{Q(E_I)}\frac{dQ(E_I)}{dE_I}\right) dE_I \tag{4.11}$$

where $E_{I(R)}(i)$ and $E_{I(R)}(i+i)$ denote the limits of the i-th bin for events with a measured ionization electron equivalent (nuclear recoil) energy.

Physically, $Q(E_R) \leq 1$ accounts for the fact that only a fraction of the recoil energy is channeled into electronic excitation (which produces the observable ionization/scintillation signal), while the remainder is lost to elastic nuclear collisions with neighboring atoms that merely displace them without producing a detectable signal. This partition between electronic and nuclear stopping was first addressed systematically by Lindhard, Scharff and Schiøtt in their study of heavy ion ranges, where they showed that at low, non-relativistic recoil velocities, both channels are comparable in magnitude and must be treated together within a unified Thomas-Fermi-like framework [44].

Following the Lindhard formalism, one introduces a dimensionless reduced energy, ϵ, that rescales E_R in terms of the target's atomic number Z and mass number A:

$$\epsilon = 11.5\, Z^{-7/3} E_R[\mathrm{keV}_{nr}], \tag{4.12}$$

in terms of which the nuclear stopping is captured by the universal function

$$g(\epsilon) = 3\epsilon^{0.15} + 0.7\epsilon^{0.6} + \epsilon, \tag{4.13}$$

and the electronic-to-nuclear stopping ratio is governed by the constant

$$k = 0.133\, Z^{2/3} A^{-1/2}, \tag{4.14}$$

which is the same proportionality constant appearing in Lindhard's electronic stopping cross section, Eq. (2.5) of [44], evaluated here for the case $Z_1 = Z_2 = Z$ (i.e. the recoiling nucleus and the stopping medium are of the same species). The quenching factor then follows as

$$Q(E_R) = \frac{k g(\epsilon)}{1 + k g(\epsilon)}. \tag{4.15}$$

For liquid argon ($Z = 18$, $A \simeq 40$), this yields $k \simeq 0.174$, consistent with values quoted in dark matter direct-detection literature that adopts the same Lindhard treatment [45].

The practical difficulty in incorporating (4.15) into the event rate calculation is that Q depends on E_R itself, so Eq. (4.10) cannot be inverted analytically to obtain $E_R(E_I)$. This inversion is nevertheless essential: experimental thresholds are set in ionization energy, $E_{I,\mathrm{th}}$, and must be translated into an equivalent recoil energy threshold, $E_{R,\mathrm{th}}$, before the lower limit of the E_R integral in (4.8) can be fixed. In practice, this is done by solving

$$E_{I,\mathrm{th}} - Q(E_R)E_R = 0 \tag{4.16}$$

numerically for E_R (e.g. via a root-finding routine such as `brentq`), since $Q(E_R)E_R$ is monotonic but non-linear over the relevant recoil range. For a typical ionization threshold of $E_{I,\mathrm{th}} \sim 100\,\mathrm{eV}_{ee}$, this procedure yields a recoil threshold of order $E_{R,\mathrm{th}} \sim 400\,\mathrm{eV}_{nr}$ in LAr.

It is worth emphasizing that the QF enters the calculation only through this threshold redefinition, and not as an additional multiplicative factor inside the integrand of (4.8): the cross section $\frac{d\sigma}{dE_R}$ remains a function of the true recoil energy throughout, and the only physical effect of quenching is to raise the lower bound of the E_R integral from a nominal (unquenched) value to $E_{R,\mathrm{th}}$ as obtained from (4.16). This distinction matters for detectors near threshold, such as the ININ TRIGA Mark III setup considered here, since a sizable fraction of the expected CEνNS spectrum lies at low recoil energies where $Q(E_R)$ changes rapidly with E_R, making the numerical inversion rather than a fixed multiplicative rescaling the correct way to propagate the quenching relation into the expected number of events.

5. Sensitivities for SBC-ININ-LAr Experiment

5.1. Introduction

This chapter presents the projected statistical sensitivity of a coherent elastic neutrino-nucleus scattering (CEνNS) experiment based on LAr target exposed to reactor antineutrinos, to two Standard Model (SM) parameters: the weak mixing angle $\sin^2\theta_W$ and the electron neutrino charge radius (NCR) $\langle r^2_{\nu_e}\rangle$. The results below correspond to different detector masses (10, 20, 100 and 200 kg), exposure times (100 and 200 days), and to the two limiting treatments of the QF discussed in the methodology chapter: an idealized QF = 1 case and the realistic case computed from the Lindhard model for argon. The chapter reproduces the projected event rates, the 1σ confidence intervals extracted from the $\Delta\chi^2$ statistic, and the corresponding $\Delta\chi^2$ curves for each configuration.

5.2. Methodology

The main results of this work are the result of calculating the statistical difference between the expected number of CEνNS events $N_{\mathrm{SM}}\left((\sin^2\theta_W)_{\mathrm{SM}}, \langle r^2_{\nu_e}\rangle_{\mathrm{SM}}\right)$ for a set value of SM parameters and a value of expected number of events obtained by varying around said parameters $N_{\mathrm{SM}\pm\delta}\left((\sin^2\theta_W)_{\mathrm{SM}} \pm \delta\left(\sin^2\theta_W\right), \langle r^2_{\nu_e}\rangle_{\mathrm{SM}} \pm \delta\langle r^2_{\nu_e}\rangle\right)$. We quantify this discrepancy through a $\Delta\chi^2$ statistic, defined as

$$\Delta\chi^2(x) = \frac{\left(N_{\mathrm{SM}} - N(x)_{\mathrm{SM}\pm\delta}\right)^2}{\sigma^2}, \tag{5.1}$$

where x denotes the parameter dependence. The variance σ^2 combines a statistical and a systematic contribution added in quadrature, following the standard treatment of counting experiments with an imperfectly known normalization. The statistical term is taken to be Poissonian, and is therefore given by the expected number of events itself, while the systematic term is assigned as a flat 10% uncertainty on the predicted event rate, chosen to conservatively account for unmodeled effects in the reactor flux normalization, the detector response, and the quenching factor treatment. Explicitly,

$$\sigma^2 = N(x)_{\mathrm{SM}} + (10\%)^2 \left(N(x)_{\mathrm{SM}\pm\delta}\right)^2, \tag{5.2}$$

where the first term on the right-hand side is the statistical (Poissonian) variance and the second term is the systematic variance, scaling quadratically with the predicted event rate as expected for a normalization-type systematic uncertainty.

5.3. Calculation Parameters

The results quoted throughout this chapter use the fixed inputs:

- $P = 1$ MW
- $d = 3$ m
- $t = 100$ or 200 days
- $M_{\text{Detector}} = 20$, 100 or 200 kg
- $N_{\text{targets}} = 3 \times 10^{26}$
- 1.5×10^{27} or 3×10^{27}
- $M = 39.95$ u $= 37.21$ GeV for the ^{40}Ar nucleus
- $M_W = 80.379$ GeV, $G_F = 1.16 \times 10^{-11}$ MeV^{-2}
- $g_n^V = -\frac{1}{2}$

together with the Standard Model reference values $\left(\sin^2\theta_W\right)_{\text{SM}} = 0.22337$ and $\langle r_{\nu_e}^2\rangle_{\text{SM}} = -0.826 \times 10^{-32}$ cm^2. The differential event rate follows the usual CEνNS convolution of the Huber-Mueller reactor flux with the CEνNS cross section, integrated over the recoil-energy window set by the ionization threshold $E_I^{\text{threshold}} \sim 100$ eV$_{\text{ee}}$, which through the Lindhard quenching relation corresponds to $E_R^{\text{threshold}} \sim 400$ eV$_{\text{nr}}$.

5.4. Projected Event Rates

Table 5.1 lists the expected number of events per day for each detector mass, separately for the two parametrizations of g_p^V (pure $\sin^2\theta_W$ dependence, and the combined $\sin^2\theta_W$-NCR dependence), and for the two QF treatments.

Table 5.1: *Projected CEνNS event rates (events/day) as a function of detector mass, for $QF = 1$ and for the Lindhard quenching model.*

g_p^V $(QF = 1)$	10 kg	20 kg	100 kg	200 kg
$\propto \sin^2\theta_W$	~ 10.1	~ 20.2	~ 101	~ 202
$\propto \sin^2\theta_W\left(1 + \frac{1}{3}M_W^2\langle r_{\nu_e}^2\rangle\right)$	~ 10.5	~ 21.1	~ 105	~ 211
g_p^V (Lindhard QF)	10 kg	20 kg	100 kg	200 kg
$\propto \sin^2\theta_W$	~ 2.2	~ 4.5	~ 22	~ 44
$\propto \sin^2\theta_W\left(1 + \frac{1}{3}M_W^2\langle r_{\nu_e}^2\rangle\right)$	~ 2.3	~ 4.7	~ 23	~ 46

The reduction of roughly a factor of 4-5 in the expected rate between the $QF = 1$ and Lindhard cases reflects the fraction of recoil events lost below the ionization threshold once the realistic quenching relation is applied, and it propagates directly into the widths of the confidence intervals discussed below.

5.5. Results

All intervals below are quoted at 68.27% confidence level (1σ), obtained from the $\Delta\chi^2$ statistic defined in Section 5.2, without including systematic correlations between $\sin^2\theta_W$ and $\langle r^2_{\nu_e}\rangle$.

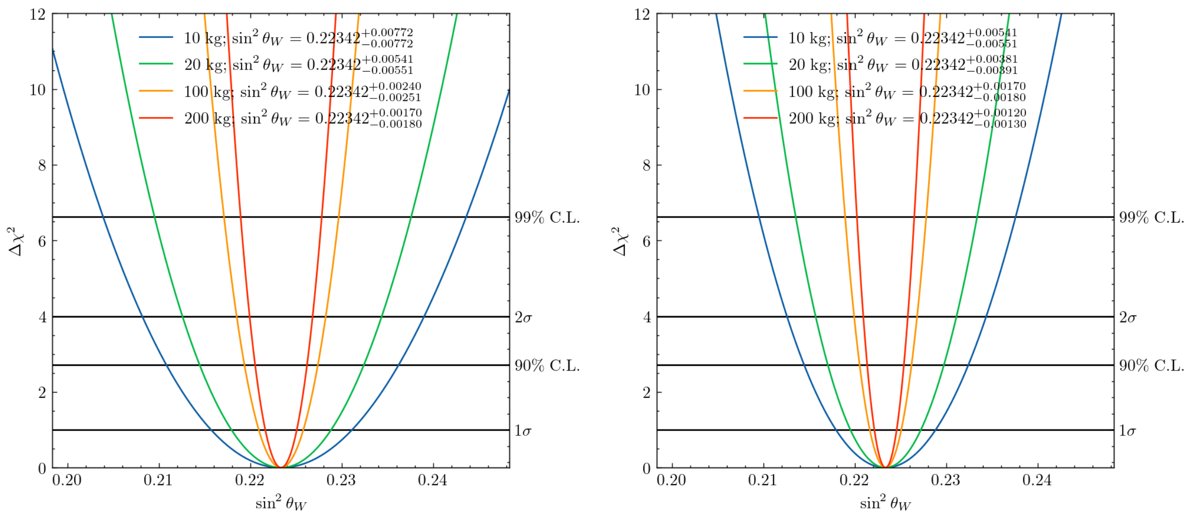
5.5.1 Without QF Influence

Sensitivity to $\sin^2\theta_W$

For an experiment with 100-day and 200-day exposure with $QF = 1$, the values at 1σ confidence level are given by Table 5.2 and the $\Delta\chi^2$ graphs are given in Figures 5.1(a) and 5.1(b):

Table 5.2: *1σ C.L. values for $\sin^2\theta_W$, without QF influence*

$\sin^2\theta_W$ ($QF = 1$)	10 kg	20 kg	100 kg	200 kg
t = 100 days	0.223 ± 0.007	0.223 ± 0.005	0.223 ± 0.003	0.223 ± 0.001
t = 200 days	0.223 ± 0.005	0.223 ± 0.003	0.223 ± 0.002	0.223 ± 0.007



(a) 100-day exposure. (b) 200-day exposure.

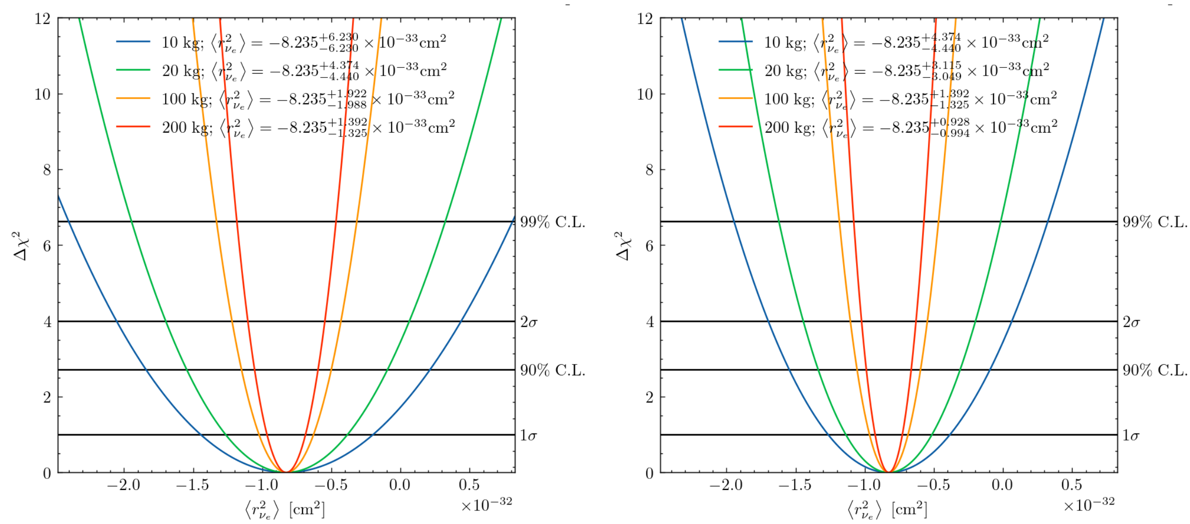
Figure 5.1: *$\Delta\chi^2$ curves for the parameter $\sin^2\theta_W$, without QF influence.*

Sensitivity to $\langle r^2_{\nu_e}\rangle$

For an experiment with 100-day and 200 day exposure with $QF = 1$, the values at 1σ confidence level are given by Table 5.3 in $\mathrm{cm}^{-2}$and the $\Delta\chi^2$ graphs are given in Figures 5.2(a) and 5.2(b):

Table 5.3: *1σ C.L. values for $\langle r^2_{\nu_e}\rangle$ in cm^2, without QF influence*

$\langle r^2_{\nu_e}\rangle$ ($QF = 1$)	10 kg	20 kg	100 kg	200 kg
t = 100 days	$-8 \pm 6 \times 10^{-33}$	$-8 \pm 4 \times 10^{-33}$	$-8 \pm 2 \times 10^{-33}$	$-8 \pm 1 \times 10^{-33}$
t = 200 days	$-8 \pm 4 \times 10^{-33}$	$-8 \pm 3 \times 10^{-33}$	$-8 \pm 1 \times 10^{-33}$	$-8.2 \pm 0.9 \times 10^{-33}$



(a) 100-day exposure. (b) 200-day exposure.

Figure 5.2: *$\Delta\chi^2$ curves for the parameter $\langle r^2_{\nu_e} \rangle$, without QF influence.*

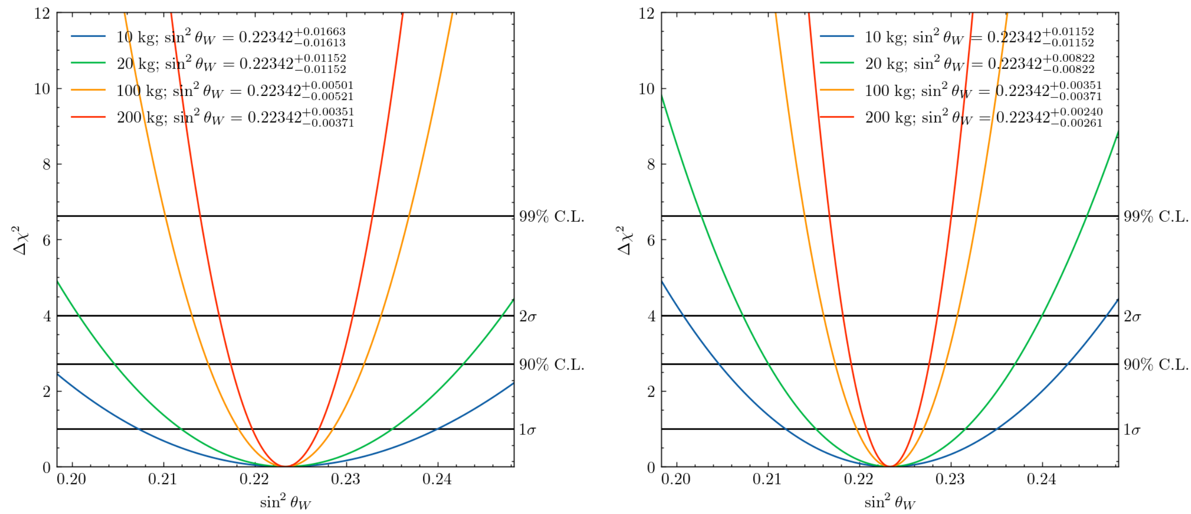
5.5.2 With QF Influence

Sensitivity to $\sin^2\theta_W$

Including the Lindhard quenching factor, the 100-day and 200-day exposure results for $\sin^2\theta_W$ yield the values shown in Table 5.4 and the $\Delta\chi^2$ graphs are shown in Figures 5.3(a) and 5.3(a).

Table 5.4: *1σ C.L. values for $\sin^2\theta_W$, with QF influence*

$\sin^2\theta_W$ (QF = Lindhard)	10 kg	20 kg	100 kg	200 kg
t = 100 days	0.22 ± 0.02	0.223 ± 0.01	0.223 ± 0.005	0.223 ± 0.003
t = 200 days	0.22 ± 0.01	0.223 ± 0.008	0.223 ± 0.003	0.223 ± 0.002



(a) 100-day exposure. (b) 200-day exposure.

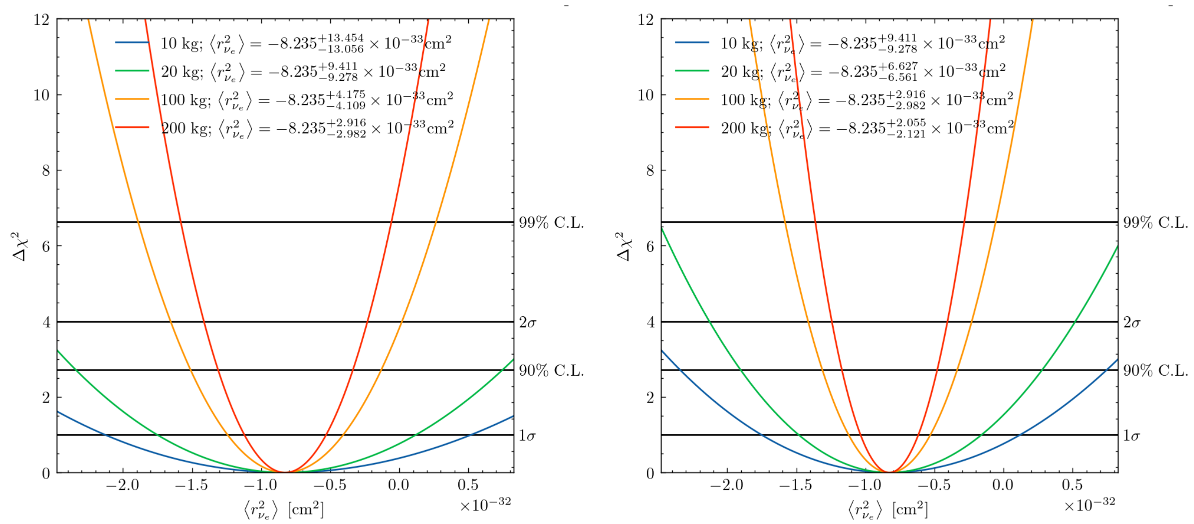
Figure 5.3: *$\Delta\chi^2$ curves for the parameter $\sin^2\theta_W$, with Lindhard QF influence.*

Sensitivity to $\langle r^2_{\nu_e} \rangle$

Including the Lindhard quenching factor, the 100-day and 200-day exposure results for $\langle r^2_{\nu_e} \rangle$ yield the values shown in Table 5.5 and the $\Delta\chi^2$ graphs are shown in Figures 5.4(a) and 5.4(b).

Table 5.5: *1σ C.L. values for $\langle r^2_{\nu_e} \rangle$ in cm^2, without QF influence*

$\langle r^2_{\nu_e} \rangle$ $(QF = 1)$	10 kg	20 kg	100 kg	200 kg
t = 100 days	$-8 \pm 6 \times 10^{-33}$	$-8 \pm 4 \times 10^{-33}$	$-8 \pm 2 \times 10^{-33}$	$-8 \pm 1 \times 10^{-33}$
t = 200 days	$-8 \pm 4 \times 10^{-33}$	$-8 \pm 3 \times 10^{-33}$	$-8 \pm 1 \times 10^{-33}$	$-8.2 \pm 0.9 \times 10^{-33}$



(a) 100-day exposure. (b) 200-day exposure.

Figure 5.4: *$\Delta\chi^2$ curves for the parameter $\langle r^2_{\nu_e} \rangle$, with Lindhard QF influence.*

6. Summary and conclusions

Coherent Elastic Neutrino-Nucleus Scattering (CEνNS) provides a uniquely clean laboratory for probing the electroweak sector of the Standard Model (SM) at low momentum transfer, since its cross section factorizes into a coherently enhanced nuclear form factor and a weak charge that carries the full dependence on the underlying couplings. In this work we exploited that sensitivity to study two related, and often degenerate, SM quantities: the weak mixing angle $\sin^2\theta_W$ and the electron neutrino charge radius (NCR) $\langle r^2_{\nu_e}\rangle$, using a liquid argon (LAr) detector exposed to the antineutrino flux of a research reactor. We began by tracing the historical and formal path that leads from Fermi's effective four-fermion interaction, through the V-A structure, the $SU(2)_L$ gauge construction, and the neutral-current and Higgs discoveries, to the modern electroweak Lagrangian, in order to motivate the neutrino electromagnetic form factors, and in particular the charge radius, as a natural and gauge-invariant extension of that framework rather than an ad hoc addition. Within that formalism we detailed the Dirac and Weyl cases for the NCR, and we clarified the numerical relation between the two conventions most commonly found in the literature, those of Bernabéu et al. and of Cadeddu et al., which differ by a factor of -2 and have historically been a source of confusion when comparing bounds across experiments.

We then derived, and made explicit, how a nonzero NCR modifies the effective proton vector coupling g_p^V that enters the CEνNS cross section, inducing a calculable shift that is degenerate, at leading order, with a shift in $\sin^2\theta_W$ itself. This degeneracy is the central technical difficulty of the analysis, and it motivated the two complementary strategies pursued in this thesis: constraining $\sin^2\theta_W$ under the assumption of a SM NCR, and constraining $\langle r^2_{\nu_e}\rangle$ under the assumption of the SM weak mixing angle, for a family of realistic experimental configurations. The event rate was computed from the convolution of the Huber-Mueller reactor antineutrino flux model with the CEνNS differential cross section on ^{40}Ar, for a detector placed 3 m from the core of the TRIGA Mark III reactor at ININ, near Mexico City, a compact and accessible neutrino source well suited to a staged, moderate-scale detector program.

A significant part of the analysis was devoted to the treatment of the detector response through the Lindhard quenching factor for argon, including the correct Jacobian treatment required to invert the relation between nuclear recoil energy and observed ionization energy. This step is not a minor technical detail: our results show that including the realistic quenching relation, rather than assuming an idealized unit response, degrades the projected precision on both $\sin^2\theta_W$ and $\langle r^2_{\nu_e}\rangle$ by a factor of two to three, since the low-energy recoils where the parameter dependence is most pronounced are precisely those most affected by quenching near threshold. We computed the projected 1σ sensitivities from a $\Delta\chi^2$ statistic, without correlations, for detector masses ranging from 10 to 200 kg and exposure times of 100 and 200 days. As expected from Poisson statistics, the sensitivity improves close to $1/\sqrt{N}$ with increasing exposure and close to an order of magnitude across the full mass range considered, with a 200 kg detector run for 200 days projected to reach a precision of order 1-2% on $\sin^2\theta_W$ and of order 2×10^{-33} cm^2 on $\langle r^2_{\nu_e}\rangle$ even under the realistic quenching scenario. These figures place a reactor-based LAr array in a competitive, if not leading, position relative to existing low-energy determinations of the weak mixing angle, and they establish a concrete, testable target for the electron neutrino charge radius using a source and detector technology already available at a national laboratory scale.

Taken together, these results confirm that a modest, single-technology detector array is sufficient to place meaningful, simultaneous constraints on $\sin^2\theta_W$ and $\langle r^2_{\nu_e}\rangle$ from reactor CEνNS, provided that the quenching response is modeled with care and that exposure and target mass are pushed toward the upper end of what is technically feasible. The analysis presented here is, by design, a first approximation: it does not yet include full covariance between $\sin^2\theta_W$ and the NCR, nor does it explore additional flavor structure or physics beyond the Standard Model. Breaking the residual degeneracy between these two parameters, and extending the sensitivity study to the muon and tau neutrino charge radii and to non-standard neutrino interactions, is the natural continuation of this program, and it is precisely the direction outlined in the doctoral protocol that follows this thesis, which proposes a systematic, multi-flavor and multi-experiment study of the neutrino charge radius across SBC, CONUS+, IsoDAR, and DUNE. Each of these experiments probes a different combination of flux, target, and baseline, and a combined treatment across them, informed by the methodology developed here for the reactor-LAr case, offers a realistic path toward disentangling the weak mixing angle from the neutrino electromagnetic structure and toward constraining genuinely new physics in the neutrino sector.

Bibliography

[1] Wolfgang Pauli. "Letter to the Physical Institute of the Federal Institute of Technology, Zürich". Open letter to the Gauverein meeting in Tübingen. Reproduced in L. M. Brown, "The idea of the neutrino," *Physics Today*, vol. 31, no. 9, pp. 23–28, 1978. Also in R. Kronig and V. F. Weisskopf (eds.), *Collected Scientific Papers by Wolfgang Pauli*, vol. 2, Interscience, New York, 1964. 1930. URL: http://www.math.utah.edu/~beebe/talks/2015/qtm/pdf/pauli-1930-ltc.pdf.

[2] C. L. Cowan et al. "Detection of the Free Neutrino: a Confirmation". In: *Science* 124.3212 (1956), pp. 103–104. DOI: 10.1126/science.124.3212.103. eprint: https://www.science.org/doi/pdf/10.1126/science.124.3212.103. URL: https://www.science.org/doi/abs/10.1126/science.124.3212.103.

[3] G. Danby et al. "Observation of High-Energy Neutrino Reactions and the Existence of Two Kinds of Neutrinos". In: *Phys. Rev. Lett.* 9 (1 1962), pp. 36–44. DOI: 10.1103/PhysRevLett.9.36. URL: https://link.aps.org/doi/10.1103/PhysRevLett.9.36.

[4] K. Kodama et al. "Observation of tau neutrino interactions". In: *Physics Letters B* 504.3 (Apr. 2001), pp. 218224. ISSN: 0370-2693. DOI: 10.1016/s0370-2693(01)00307-0. URL: http://dx.doi.org/10.1016/S0370-2693(01)00307-0.

[5] K. Kodama et al. "Final tau-neutrino results from the DONuT experiment". In: *Phys. Rev. D* 78 (5 2008), p. 052002. DOI: 10.1103/PhysRevD.78.052002. URL: https://link.aps.org/doi/10.1103/PhysRevD.78.052002.

[6] O. Klein and Y. Nishina. "Über die Streuung von Strahlung durch freie Elektronen nach der neuen relativistischen Quantendynamik von Dirac". In: *Zeitschrift für Physik* 52.11-12 (Nov. 1929), pp. 853868. ISSN: 0044-3328. DOI: 10.1007/bf01366453. URL: http://dx.doi.org/10.1007/BF01366453.

[7] C. Møller. "Zur Theorie des Durchgangs schneller Elektronen durch Materie". In: *Annalen der Physik* 406.5 (1932), pp. 531–585. DOI: https://doi.org/10.1002/andp.19324060506. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/andp.19324060506. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/andp.19324060506.

[8] T. D. Lee and C. N. Yang. "Question of Parity Conservation in Weak Interactions". In: *Phys. Rev.* 104 (1956), pp. 254–258. DOI: 10.1103/PhysRev.104.254.

[9] C. S. Wu et al. "Experimental Test of Parity Conservation in Beta Decay". In: *Phys. Rev.* 105 (1957), pp. 1413–1415. DOI: 10.1103/PhysRev.105.1413.

[10] E. C. G. Sudarshan and R. E. Marshak. "Chirality Invariance and the Universal Fermi Interaction". In: *Phys. Rev.* 109 (1958), pp. 1860–1862. DOI: 10.1103/PhysRev.109.1860.

[11] R. P. Feynman and M. Gell-Mann. "Theory of the Fermi Interaction". In: *Phys. Rev.* 109 (1958), pp. 193–198. DOI: 10.1103/PhysRev.109.193.

[12] N. Cabibbo. "Unitary Symmetry and Leptonic Decays". In: *Phys. Rev. Lett.* 10 (1963), pp. 531–533. DOI: 10.1103/PhysRevLett.10.531.

[13] S. L. Glashow, J. Iliopoulos, and L. Maiani. "Weak Interactions with Lepton-Hadron Symmetry". In: *Phys. Rev. D* 2 (1970), pp. 1285–1292. DOI: 10.1103/PhysRevD.2.1285.

[14] Sheldon L. Glashow. "Partial-symmetries of weak interactions". In: *Nuclear Physics* 22.4 (1961), pp. 579–588. ISSN: 0029-5582. DOI: https://doi.org/10.1016/0029-5582(61)90469-2. URL: https://www.sciencedirect.com/science/article/pii/0029558261904692.

[15] F.J. Hasert et al. "Search for elastic muon-neutrino electron scattering". In: *Physics Letters B* 46.1 (1973), pp. 121–124. ISSN: 0370-2693. DOI: https://doi.org/10.1016/0370-2693(73)90494-2. URL: https://www.sciencedirect.com/science/article/pii/0370269373904942.

[16] F.J. Hasert et al. "Observation of neutrino-like interactions without muon or electron in the gargamelle neutrino experiment". In: *Physics Letters B* 46.1 (1973), pp. 138–140. ISSN: 0370-2693. DOI: https://doi.org/10.1016/0370-2693(73)90499-1. URL: https://www.sciencedirect.com/science/article/pii/0370269373904991.

[17] Steven Weinberg. "A Model of Leptons". In: *Phys. Rev. Lett.* 19 (21 1967), pp. 1264–1266. DOI: 10.1103/PhysRevLett.19.1264. URL: https://link.aps.org/doi/10.1103/PhysRevLett.19.1264.

[18] Abdus Salam and J. C. Ward. "Weak and electromagnetic interactions". In: *Il Nuovo Cimento* 11.4 (Feb. 1959), pp. 568577. ISSN: 1827-6121. DOI: 10.1007/bf02726525. URL: http://dx.doi.org/10.1007/BF02726525.

[19] Peter W. Higgs. "Broken Symmetries and the Masses of Gauge Bosons". In: *Phys. Rev. Lett.* 13 (16 1964), pp. 508–509. DOI: 10.1103/PhysRevLett.13.508. URL: https://link.aps.org/doi/10.1103/PhysRevLett.13.508.

[20] F. Englert and R. Brout. "Broken Symmetry and the Mass of Gauge Vector Mesons". In: *Phys. Rev. Lett.* 13 (9 1964), pp. 321–323. DOI: 10.1103/PhysRevLett.13.321. URL: https://link.aps.org/doi/10.1103/PhysRevLett.13.321.

[21] José F. Nieves. "Electromagnetic properties of Majorana neutrinos". In: *Phys. Rev. D* 26 (11 1982), pp. 3152–3158. DOI: 10.1103/PhysRevD.26.3152. URL: https://link.aps.org/doi/10.1103/PhysRevD.26.3152.

[22] Boris Kayser. "Majorana neutrinos and their electromagnetic properties". In: *Phys. Rev. D* 26 (7 1982), pp. 1662–1670. DOI: 10.1103/PhysRevD.26.1662. URL: https://link.aps.org/doi/10.1103/PhysRevD.26.1662.

[23] Carlo Giunti and Alexander Studenikin. "Neutrino electromagnetic interactions: A window to new physics". In: *Rev. Mod. Phys.* 87 (2 2015), pp. 531–591. DOI: 10.1103/RevModPhys.87.531. URL: https://link.aps.org/doi/10.1103/RevModPhys.87.531.

[24] M Nowakowski, E A Paschos, and J M Rodríguez. "All electromagnetic form factors". In: *European Journal of Physics* 26.4 (2005), pp. 545–560. ISSN: 1361-6404. DOI: 10.1088/0143-0807/26/4/001. URL: http://dx.doi.org/10.1088/0143-0807/26/4/001.

[25] Ziro Maki, Masami Nakagawa, and Shoichi Sakata. "Remarks on the Unified Model of Elementary Particles". In: *Progress of Theoretical Physics* 28.5 (Nov. 1962), pp. 870–880. ISSN: 0033-068X. DOI: 10.1143/PTP.28.870. eprint: https://academic.oup.com/ptp/article-pdf/28/5/870/5258750/28-5-870.pdf. URL: https://doi.org/10.1143/PTP.28.870.

[26] J W F Valle. "Neutrino physics overview". In: *Journal of Physics: Conference Series* 53.1 (2006), p. 473. DOI: 10.1088/1742-6596/53/1/031. URL: https://doi.org/10.1088/1742-6596/53/1/031.

[27] J. Bernabéu et al. "Charge radius of the neutrino". In: *Phys. Rev. D* 62 (11 2000), p. 113012. DOI: 10.1103/PhysRevD.62.113012. URL: https://link.aps.org/doi/10.1103/PhysRevD.62.113012.

[28] Daniel Z. Freedman. "Coherent effects of a weak neutral current". In: *Phys. Rev. D* 9 (5 1974), pp. 1389–1392. DOI: 10.1103/PhysRevD.9.1389. URL: https://link.aps.org/doi/10.1103/PhysRevD.9.1389.

[29] D. Akimov et al. "Observation of coherent elastic neutrino-nucleus scattering". In: *Science* 357.6356 (2017), pp. 1123–1126. DOI: 10.1126/science.aao0990. eprint: https://www.science.org/doi/pdf/10.1126/science.aao0990. URL: https://www.science.org/doi/abs/10.1126/science.aao0990.

[30] Kate Scholberg. "Coherent Elastic Neutrino-Nucleus Scattering". In: *Journal of Physics: Conference Series* 1468.1 (2020), p. 012126. DOI: 10.1088/1742-6596/1468/1/012126. URL: https://dx.doi.org/10.1088/1742-6596/1468/1/012126.

[31] D. K. Papoulias and T. S. Kosmas. "Standard and Nonstandard Neutrino-Nucleus Reactions Cross Sections and Event Rates to Neutrino Detection Experiments". In: *Advances in High Energy Physics* 2015 (2015), pp. 117. ISSN: 1687-7365. DOI: 10.1155/2015/763648. URL: http://dx.doi.org/10.1155/2015/763648.

[32] Arie Bodek, Hyon-San Seo, and Un-Ki Yang. *Precision Measurements of the Electroweak Mixing Angle in the Region of the Z pole*. 2025. arXiv: 2508.18022 [hep-ex]. URL: https://arxiv.org/abs/2508.18022.

[33] P. Aliani et al. "Neutrino mass parameters from Kamland, SNO, and other solar evidence". In: *Physical Review D* 69.1 (2004). DOI: 10.1103/physrevd.69.013005. URL: https://doi.org/10.1103%2Fphysrevd.69.013005.

[34] Daya Bay Collaboration. *A Precision Measurement of the Neutrino Mixing Angle θ_{13} using Reactor Antineutrinos at Daya Bay*. 2007. arXiv: hep-ex/0701029 [hep-ex].

[35] L. J. Flores et al. "Physics reach of a low threshold scintillating argon bubble chamber in coherent elastic neutrino-nucleus scattering reactor experiments". In: *Physical Review D* 103.9 (May 2021). ISSN: 2470-0029. DOI: 10.1103/physrevd.103.l091301. URL: http://dx.doi.org/10.1103/PhysRevD.103.L091301.

[36] E. Alfonso-Pita et al. "New physics searches in a low threshold scintillating argon bubble chamber measuring coherent elastic neutrino-nucleus scattering in reactors". In: *Physical Review D* 105.11 (2022). ISSN: 2470-0029. DOI: 10.1103/physrevd.105.113005. URL: http://dx.doi.org/10.1103/PhysRevD.105.113005.

[37] F. von Feilitzsch, A.A. Hahn, and K. Schreckenbach. "Experimental beta-spectra from 239Pu and 235U thermal neutron fission products and their correlated antineutrino spectra". In: *Physics Letters B* 118.1 (1982), pp. 162–166. ISSN: 0370-2693. DOI: https://doi.org/10.1016/0370-2693(82)90622-0. URL: https://www.sciencedirect.com/science/article/pii/0370269382906220.

[38] K. Schreckenbach et al. "Determination of the antineutrino spectrum from 235U thermal neutron fission products up to 9.5 MeV". In: *Physics Letters B* 160.4 (1985), pp. 325–330. ISSN: 0370-2693. DOI: https://doi.org/10.1016/0370-2693(85)91337-1. URL: https://www.sciencedirect.com/science/article/pii/0370269385913371.

[39] Patrick Huber and Thomas Schwetz. "Precision spectroscopy with reactor antineutrinos". In: *Phys. Rev. D* 70 (5 2004), p. 053011. DOI: 10.1103/PhysRevD.70.053011. URL: https://link.aps.org/doi/10.1103/PhysRevD.70.053011.

[40] Th. A. Mueller et al. "Improved predictions of reactor antineutrino spectra". In: *Phys. Rev. C* 83 (5 2011), p. 054615. DOI: 10.1103/PhysRevC.83.054615. URL: https://link.aps.org/doi/10.1103/PhysRevC.83.054615.

[41] V. I. Kopeikin, L. A. Mikaelyan, and V. V. Sinev. "Reactor as a source of antineutrinos: Thermal fission energy". In: *Physics of Atomic Nuclei* 67.10 (2004), pp. 1892–1899. ISSN: 1562-692X. DOI: 10.1134/1.1811196. URL: https://doi.org/10.1134/1.1811196.

[42] H. T. Wong et al. "Search of neutrino magnetic moments with a high-purity germanium detector at the Kuo-Sheng nuclear power station". In: *Phys. Rev. D* 75 (1 2007), p. 012001. DOI: 10.1103/PhysRevD.75.012001. URL: https://link.aps.org/doi/10.1103/PhysRevD.75.012001.

[43] Guillermo Fernandez-Moroni et al. "The physics potential of a reactor neutrino experiment with Skipper CCDs: measuring the weak mixing angle". In: *Journal of High Energy Physics* 2021.3 (Mar. 2021). ISSN: 1029-8479. DOI: 10.1007/jhep03(2021)186. URL: http://dx.doi.org/10.1007/JHEP03(2021)186.

[44] J Lindhard, M Scharff, and H E Schiø tt. "RANGE CONCEPTS AND HEAVY ION RANGES (NOTES ON ATOMIC COLLISIONS, II)". In: *Kgl. Danske Videnskab. Selskab. Mat. Fys. Medd.* Vol: 33: No. 14 (Jan. 1963). URL: https://www.osti.gov/biblio/4153115.

[45] J.D. Lewin and P.F. Smith. "Review of mathematics, numerical factors, and corrections for dark matter experiments based on elastic nuclear recoil". In: *Astroparticle Physics* 6.1 (1996), pp. 87–112. ISSN: 0927-6505. DOI: https://doi.org/10.1016/S0927-6505(96)00047-3. URL: https://www.sciencedirect.com/science/article/pii/S0927650596000473.